The Journal of Systems and Software 226 (2025) 112407

Contents lists available at ScienceDirect

The Journal of Systems & Software

journal homepage: www.elsevier.com/locate/jss

Requirements extraction from model-based systems engineering: A systematic literature review ☆

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ARTICLE INFO

Dataset link: www.zenodo.org/

Keywords:
Model-based systems engineering
Requirements engineering
Requirements extraction
Systematic literature review
Systems engineering

ABSTRACT

Collaboration and easy data exchange are crucial in modern systems that involve hardware, electronics, software, and users. Requirement Engineering (RE) and Systems Engineering (SE) are challenging fields that require tool support to automate activities. Natural language (NL) requirement documents can create processing issues. To address these issues, detailed models have been developed to represent a system effectively. These models are intend to replace inconsistent documents over time by using model-based methodologies like Model-Based SE (MBSE). Within the MBSE methodologies, Arcadia/Capella has proven its capabilities as a comprehensive tool in the SE community to define and validate complex system architecture. Thus, this paper aims to investigate the tools, methods, techniques, or processes for extracting requirements from the MBSE environment or model generation from NL requirements. Furthermore, this discusses how these approaches are applied specifically in the Arcadia/Capella and how transforming requirements are addressed to support textual requirements. We conducted a systematic literature review (SLR) by selecting 97 articles to examine advances in this field in various aspects of these approaches. The results presented in this SLR uncovered several key findings that have important implications for future research, such as the dominance of the model generation from NL; transforming model-based requirements to NL requires more data; and the fact that requirements extraction in Arcadia/Capella needs more evidence.

1. Introduction

Modern systems are becoming increasingly complex, driven by the interaction of hardware, electronics, software, and users. System designs often involve collaboration across multiple organizations and require easy data exchange (Kossiakoff et al., 2020). In this context, RE has become a challenging field in system development and is crucial for development success (Nuseibeh and Easterbrook, 2000). The RE process emphasizes systematic techniques that ensure completeness, consistency, and consequently minimize errors (Geogy and Dharani, 2016). RE involves capturing and analyzing stakeholder needs, and NL documents serve as one compelling medium for expressing and documenting those requirements (Umar and Lano, 2024).

However, large requirement documents can cause issues during processing (Ahmed et al., 2022), which industry experts and the research community recognize as a need for tool support to automate RE activities (Umar and Lano, 2024). Automating RE processes can help bridge the gap caused by missing and omitted functional requirements identified during the RE specification phase. RE automation is the final step after addressing all requirement iterations and associations during requirement modeling, which is sometimes represented in textual documentation. Employing the concept of ontology has been used to specify complete and unambiguous requirements (Castañeda et al., 2010; Siegemund et al., 2011), and leveraging techniques such as Natural Language Processing (NLP) can improve the effectiveness and efficacy of the RE process and in dealing with NL artifacts (Umar and Lano, 2024).

For the reasons cited above, the SE community has resorted to the assistance of an integrated set of detailed descriptive models to represent a system. These models have become the main means of capturing and communicating knowledge about a system over its lifetime, replacing many discrete documents that have become inconsistent over time (Noguchi, 2016). The process entails specific requirements articulated as model-based requirements and supported by methodologies such as MBSE.

MBSE is an emerging approach in the SE field (Rhodes, 2008; Grady, 2009) that supports the requirements definition, design, analysis, verification, and validation stages of the system under development

☆ Editor: Heiko Koziolek.
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https://doi.org/10.1016/j.jss.2025.112407
Received 12 May 2024; Received in revised form 23 December 2024; Accepted 20 February 2025
Available online 4 March 2025

from the beginning of the conceptual design phase and is continuously used throughout the other phases of the lifecycle (Friedenthal et al., 2014). Combined with the advent of formal modeling languages such as Unified Modeling Language (UML) and Systems Modeling Language (SysML), MBSE methodology allows systems engineers to create high-fidelity representations of system requirements, behaviors, and structures (Kossiakoff et al., 2020).

Within the MBSE methodologies (Di Maio et al., 2021), the Arcadia method was developed by Thales Company and consists of a set of processes, methods, and tools (Estefan et al., 2007) with a focus on defining and validating the architecture of complex systems. Also, it is a methodology that provides both a method and a language, adopting concepts inspired by UML and SysML, using Capella software as a tool. Moreover, Capella stands out by allowing the introduction of logical functions into the logical architecture. This unique feature eases appropriation and understanding by all stakeholders, setting it apart from other SysML tools. These attributes have made Capella a comprehensive tool in the SE community.

However, unlike SysML, Capella does not feature any specific modeling notation to describe requirements textually. Still, it allows one to create requirements elements, linking them to the model elements in any diagram (Cunha and Cerqueira, 2023). In this context, applying a plugin or add-on to extract the functional requirements defined as model elements and their relationships and iterations in Capella into textual requirements would represent a significant benefit for the SE community.

Grounded on the immense potential of technologies used to extract text-based requirements from the MBSE environment or model generation from NL requirements, we conducted an SLR to examine the advancements achieved in this field in various aspects of these approaches. Our main objective is to investigate the tools, methods, techniques, or processes to transform requirements from or to an MBSE environment. Additionally, how these approaches are applied in SE, the types of requirements model, in which RE phases are used, and the kinds of transformation applied.

Also, we intend to identify how requirement extraction is applied to support textual requirements, the model diagram involved, and the transformation results used. Furthermore, we want to investigate how requirement extraction is applied in the Arcadia method, analyzing the model diagrams, relationships, iterations involved, and ways to export to a specification form. Finally, our objective is to verify the potential risks and challenges that can help objectively assess research.

The motivation for this study arises from the gap between the conceptualized product in a large NL requirement document by the system architecture and the product designed by the development team. Specifically, we aim to examine the impact of the automated functional requirements generation from models to mitigate missing requirements and add completeness to the RE process, besides observing the incremental value in both conception and execution in the development of systems in MBSE.

The study's contribution can be summarized as follows: (i) Extensive inquiry: This SLR examines various aspects of extracting or transforming requirements within the SE domain. In addition, it analyzes the relations to MBSE, textual requirements, the Arcadia method, and the methodologies used to evaluate these approaches. (ii) Quality issues: the study is motivated by the influence of automating functional requirements toward completeness. Completeness is a crucial attribute for high-quality requirements and specifications. Missing and incomplete requirements can be mitigated by addressing functions, components, and iterations within an optimized approach to the requirements specification (iii) Reducing cost: the potential impact is substantial given the cost generated to corporations, especially if requirements errors are only identified in the final stages of development.

We organized this paper as follows. Section 2 provides an overview of the technical background and related work. Section 3 outlines the research methodology utilized for conducting the SLR. Section 4 presents the results and analysis concerning our research questions (RQ). Our conclusions are provided in Section 5.

2. Background and related work

2.1. Definitions

We present the following definitions alphabetically to ensure consistency throughout this paper and set the scope of the SLR.

Approach: In the context of this SLR, we are interested in specific approaches used for requirements extraction, such as techniques, models, frameworks, methods, processes, and methodologies.

Arcadia: It is a structured engineering method to define and verify the architecture of complex systems, providing a means to carry out, already in the definition phase, the iterations that will allow the convergence of the architecture for the adequacy of all identified requirements to the subsystems, integration, verification, and validation phases (Voirin, 2018). The Arcadia method offers a straightforward approach to managing requirements by embedding them within the system models. However, it is important to note that it does not eliminate the need for dedicated requirements management software. Instead, Arcadia focuses on establishing connections (Roques, 2017) between requirements and various model elements, such as functions, functional chains, and capabilities. This means that it helps organize and relate requirements to different system components. However, it might still need some of the comprehensive features specialized software provides for requirements management.

Information extraction: The extraction process is a fundamental aspect of NLP, which aims to derive insights and information from textual data. In the context of this SLR, the objective is to develop a system that can comprehend the meaning of statements and models, including the contextual subtleties of the language used. Several NLP techniques can extract information from documents, such as entity names, locations, quantities, etc. Additionally, other methods can extract information related to associations, inheritance, and aggregation through patterns identification.

MBSE: It is a technical approach to SE that focuses on creating and exploiting domain models as the primary means of information exchange rather than document-based. It can also be described as the formalized application of modeling principles, methods, languages, and tools to the entire lifecycle of large, complex, interdisciplinary, socio-technical systems (Ramos et al., 2011). The purpose of MBSE, as described by Noguchi (2016), is to enhance the efficiency and effectiveness of SE by utilizing an integrated descriptive representation of the system to capture knowledge about the system for the benefit of all stakeholders.

Model-based requirement: It refers to using detailed models instead of textual requirements to define, analyze, and manage system requirements. This improves the understanding of system behavior and structure through their interactions (Sommerville, 2011). The models are utilized to support the entire lifecycle of a system, including design, analysis, verification, and validation phases.

NLP: It is a field of study that combines computer science and computational linguistics. Its main goal is to enable computers to interact with and understand human language. It involves the development of algorithms and techniques that allow computers to process, interpret, and generate human language data in a meaningful and contextually relevant way. NLP covers many tasks, including text processing, speech processing, syntactic analysis, and morphological analysis. Details of these techniques may be found in Bird et al. (2009).

Requirement: According to Institute of Electrical and Eletronics Engineers (1990), there are three instances in which the requirements definitions are established. They are as follows: (i) a condition or capability needed by a user to solve a problem or achieve an objective; (ii) a condition or capability that must be met or possessed by a system or system component to satisfy a contract, standard, specification, or other formally imposed documents; (iii) a documented representation of a condition or capability as in (i) or (ii).

SysML: It is the most widely used MBSE language. The Object Management Group (OMG) developed it in partnership with International Council on SE (INCOSE) for SE applications. It supports the RE phases, such as specification, analysis, design, verification, and validation of a wide range of systems and systems-of-systems. SysML is defined as an extension of the UML using UML's profile mechanism. According to Kossiakoff et al. (2020), its strength is its usefulness at multiple levels of abstraction, from high-level concept modeling to the extreme detail required to model pinouts in a physical architecture.

UML: It is a general-purpose visual modeling language intended to provide a standard way to visualize a system's design. It was developed to help system and software developers specify, visualize, construct, and document software system artifacts. UML provides a standard notation for many types of diagrams (France et al., 1998), which can be divided into two main groups: behavior and structure diagrams.

2.2. Related work

Loniewski et al. (2010) conducted an SLR to examine the use of RE techniques in Model-Driven Development (MDD) and their level of automation. According to the author, the main objective is to develop a process for gathering current knowledge about RE techniques in MDD and identifying research gaps. The study is focused on research for automating model transformations in the requirements phase, demonstrating simple approaches without explicitly showing any variation with a specific attribute such as RE modeling styles.

Umar and Lano (2024) conducted, in 2023, an SLR to examine empirical evidence on automated RE, aiming at their impact on software development. The authors explored various dimensions of automated RE tools, such as the output, the RE phases applied, and the automation degree. The study obtained relevant responses related to our SLR theme as follows: UML was the most output supported by automated tools; most automated tools are developed to automate activities in the RE analysis phase; and NLP was responsible for 50,6% of the technologies employed in the context of the automated RE tools. The study differs from ours with regard to the need for additional information related explicitly to MBSE. In general, it emphasizes automated RE tools.

Yue et al. (2011b) conducted an SLR of transformation approaches between user requirements and analysis models. The study aimed to examine existing literature works that transform textual requirements into analysis models and highlight open issues. The study presented detailed content focused on the transformation approach but was not directly applicable to requirement extraction from MBSE.

Ahmed et al. (2022) conducted an SLR of automatic transformation of NL to UML, which aimed to understand existing systems' effectiveness better and provide a conceptual framework with further improvement guidelines. The study presented statistics related to approaches to convert NL to UML and automation to convert NL to UML. Moreover, this study presented relationships among UML components, the distribution of final/intermediate outputs acquired, and finally, the usage of technologies or tools to solve the problems. The study's topic partially covers our theme but does not address the transformation of UML to NL.

Nicolás and Toval (2009) conducted an SLR on the generation of textual requirements specifications from software engineering models. The study aimed to review the combination of software engineering models and textual requirements in the requirements specification process, particularly those approaches that generate text from models. The study focused on finding methods and techniques for generating, translating, combining, integrating, or synchronizing models and textual requirements, in this order: from models to requirements. The study restricted their findings to the field of SE, clearly focusing Software Product Line (SPL) models. Despite this, it is worth highlighting the findings from models to requirements specification, which are relevant to our SLR topic.

Wolny et al. (2020) conducted a systematic mapping study that examines the usage and application of SysML during its initial thirteen years. Their analysis revealed some findings such as: there is a growing scientific interest in SysML, mainly in the research field of SE; SysML is primarily used in the design or validation phase rather than in the implementation phase; the most commonly used diagram types are the SysML-specific requirement diagram, parametric diagram, and block diagram, together with the activity diagram and state machine diagram known from UML; SysML is a specific UML profile primarily used in SE; however, the language has to be customized to accommodate domain-specific aspects. The article sparked our interest in investigating our SLR subject; however, it did not provide any evidence that would compare to ours.

Dermeval et al. (2016) conducted an SLR to explore the use of ontologies in RE. This publication aimed to investigate how ontologies support RE activities. Even being presented as a topic not primarily connected with this SLR, the study complements our addressed research subjects, such as the RE modeling styles and the RE phases in which the ontologies were employed.

Yang et al. (2014) conducted an SLR of requirements modeling and analysis for a self-adaptive system. The study aimed to investigate what modeling methods, RE activities, requirements quality attributes, application domains, and research topics have been studied and how well these studies have been conveyed. The referenced SLR presents a comprehensive overview of RE modeling besides relevant topics to our study, such as modeling methods, on the other hand, it is important to cite that it does not address the aspects of requirement transformation or extraction.

Teufl et al. (2013) conducted an SLR focused on the requirements for a model-based RE tool specifically designed for embedded systems. The study was motivated by the lack of appropriate tools that support model-based development. Besides the domain application, the study differs from ours regarding the quality aspects of RE, such as checking the correctness and relevance of each requirement and validating completeness.

While the related work provides a comprehensive discussion on requirements extraction and modeling techniques, it does not address the specific research gap this SLR aims to fill. Our aim is to explore the advancements in requirements extraction within MBSE, with a particular focus on the Arcadia/Capella methodology, and identifying potential areas for future research to enhancement of these processes.

3. Research methodology

This section presents the methodology adopted in this SLR, according to the guidelines (Kitchenham et al., 2007, 2015; Biolchini et al., 2005). Fig. 1 illustrates our 12-step process, divided into planning, conducting, and reporting the SLR. Section 1 already discussed the demand of SLR (first step in our process), while the Sections 3.2 to 3.6 outline the remaining steps.

We used the Parsifal[1] tool to document our SLR protocol (steps 2 to 4) and carry out the process (steps 5 to 9). The tool allows for multiple users collaborating in the SLR, and studies have shown that using such a tool has positive results for executing SLRs (Stefanovic et al., 2021).

In Step 3 - Define a Research Question, we formulated a primary research question (see RQ1, in Table 1) focused on the central theme of this SLR. We have also developed complementary RQs (RQ2, RQ3, and RQ4) to explore the contexts in which these methods were applied. Finally, we added RQ5 to examine the benefits and challenges associated with their use in RE. After establishing these RQs, we applied the PICOC framework (which stands for Population, Intervention, Comparison, Outcome, and Context (Wohlin et al., 2012)) to develop a search string aligned with these components:

[1] https://parsif.al/.

Table 1
Research questions.

ID	Research questions	Objective/Aim
RQ1	What are the tools, methods, techniques, or processes to extract requirements from an MBSE environment or model generation from NL requirements?	Identify tools, methods, techniques, processes, and applications for extracting text-based requirements in an MBSE environment.
RQ2	How are these tools identified in RQ1 applied in SE?	Identify how requirements extraction is applied in SE, the types of requirements model, in which phases they are used, and the kinds of transformation applied.
RQ3	How are these tools identified in RQ1 applied to support textual requirements?	Identify how requirement extraction is applied to support textual requirements, the model diagrams involved, and the transformation results were applied.
RQ4	How are these tools identified in RQ1 applied in the Arcadia method?	Identify how requirements extraction is applied in the Arcadia method, analyze the model diagrams, relationships, iterations involved, and ways to export to a specification form.
RQ5	What are the benefits and difficulties in obtaining the requirements?	Identify the potential risks and challenges to help in making an objective assessment of the research.

Table 2
Search string for each electronic database.

Database	Search engine URL	Search string	Results
ACM	https://dl.acm.org/search/advanced	`((Requirement* AND (tool* OR application* OR methodolog* OR process* OR technique* OR model* OR framework OR software) AND (Arcadia OR MBSE OR SysML OR UML) AND ("model-based requirements" OR "model-based requirement representations" OR "natural language processing" OR "requirements model"))`	16
IEEE Xplore	https://ieeexplore.ieee.org/Xplore	`(Requirement AND (tool OR application OR methodolog OR process OR technique OR model OR framework OR software) AND (Arcadia OR MBSE OR SysML OR UML) AND ("model-based requirements" OR "model-based requirement representations" OR "natural language processing" OR "requirements model"))`	123
Emerald	https://www.emerald.com/insight/	`(Requirement* AND (tool* OR application* OR methodolog* OR process* OR technique* OR model* OR framework OR software) AND (Arcadia OR MBSE OR SysML OR UML) AND ("model-based requirements" OR "model-based requirement representations" OR "natural language processing" OR "requirements model"))`	71
ScienceDirect	https://www.sciencedirect.com/	`(Arcadia OR MBSE OR SysML OR UML) AND ("model-based requirements" OR "model-based requirement representations" OR "natural language processing" OR "requirements model"))`	93
Springer	https://link.springer.com/	`(Requirement* AND (tool* OR application* OR methodolog* OR process* OR technique* OR model* OR framework OR software) AND (Arcadia OR MBSE OR SysML OR UML) AND ("model-based requirements" OR "model-based requirement representations" OR "natural language processing" OR "requirements model"))`	761

- **Population:** requirement engineer, system engineer and development engineer
- **Intervention:** tools, applications and methodologies to extract engineering requirements.
- **Comparison:** Arcadia, MBSE, SysML, UML
- **Outcome:** techniques to extract functional requirements in relation of consistency and completeness
- **Context:** RE during system engineering modeling

3.1. Search strategy

3.1.1. Preliminary search

Our search strategy consisted primarily of electronic databases. We started with a preliminary search using Google Scholar to gain an initial understanding of the available literature and databases indexing them. We formulated a generic search string `(Extracting AND/OR Elaborating) AND Requirement (UML AND/OR SysML AND/OR MBSE) AND (Review OR SLR OR Research overview)` and conducted a trial search to identify relevant studies, yielding 639 results.

3.1.2. Database search

Given the lack of necessary features for SLRs and the reported low precision of Google Scholar (Boeker et al., 2013), we transitioned to more specialized databases. The results of our preliminary search were the basis for selecting electronic databases and formulating specific search strings. These search strings were carefully formulated to ensure the studies retrieved were pertinent to the research topic.

Each search engine required a tailored strategy, including methods such as filtering by title, abstract, full content, and the use of operators and wildcards. We performed two rounds of tests in each search engine to limit the number of studies and achieve a particular relevance of the articles returned by the search, except for Springer, where we

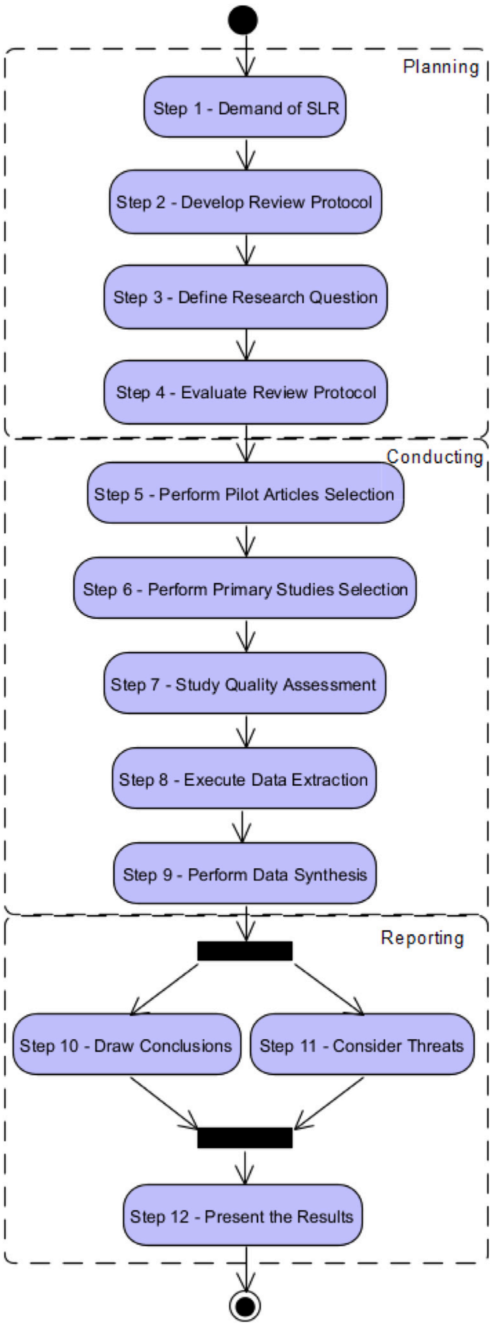

Fig. 1. SLR steps adapted from Kitchenham et al. (2007, 2015), Biolchini et al. (2005).

performed three rounds for the same purpose. Each round of test involved verifying whether the abstracts of 20 first results adhered to the theme of this SLR. Finally, we retrieved a total of 1064 studies from database searches. Table 2 provides a list of the five databases, their search engine, specific search strings, and search results.

3.1.3. Snowballing

After study selection (see Section 3.2 below), we noted an absence of sources directly addressing RQ4, i.e. specifically targeted at the Arcadia method. Given that this RQ pertains to a relatively new topic, and to identify additional evidence not indexed in the electronic databases, we used the snowballing approach (Wohlin, 2014) to uncover further relevant studies.

We conducted a forward snowballing using two key books on the Arcadia method cited on the Capella website[2]: (Voirin, 2018; Roques, 2017) as the starting seed. This approach led to the identify 204 additional papers, of which, 121 derived from Voirin (2018) and 83 from Roques (2017). These candidate papers were also subjected to our selection process; however, none met the inclusion criteria. As a result, we were unable to find sufficient evidence to adequately address RQ4.

3.2. Study selection

3.2.1. Pilot selection criteria

As part of our review protocol (step 2 in Fig. 1), we drafted selection criteria to assess whether a candidate paper is relevant to our RQs. To be included, studies must meet all of the following conditions:

- I1 peer-reviewed studies of the scientific community;
- I2 published after the year 2000, as this marks the first advancements in RE and MBSE;
- I3 articles from journals, conferences, and magazines;
- I4 addressing model-based requirements between the requirements analysis and specification phases;
- I5 addressing NLP or ontologies aimed at modeling requirements;
- I6 focusing on tools, methods, techniques, or processes for extracting requirements from an MBSE environment or model generation from NL requirements; and
- I7 meeting a minimum quality threshold (see Section 3.3).

We excluded studies that met at least one of the following exclusion criteria:

- E1 non-peer-reviewed studies;
- E2 duplicate studies;
- E3 studies not written in English;
- E4 short papers and conference proceedings;
- E5 studies not related to RE between analysis and specification phases;
- E6 studies unrelated to MBSE environmental and do not focus on RE, such as studies that address web applications or data warehouse design;
- E7 studies that do not discuss tools, methods, techniques, or processes for extracting requirements or using ontology-driven approaches to support RE; and
- E8 secondary or review papers that discuss many tools simultaneously.

3.2.2. Pilot selection

We conducted an initial pilot selection (Step 5) of 20 random articles to assess the consistency of the selection criteria between two independent researchers. Using Cohen's Kappa Coefficient (Landis and Koch, 1977), we computed a moderate level of inter-rater agreement (k=0.444). After this initial check, we made changes to the criteria to better target papers focused on requirement extraction. In a second pilot round, we assess another sample of 20 articles to check if there was an improvement in the inter-rater agreement compared to the first round. We improved the level of agreement to substantial (k=0.667).

3.2.3. Selection process

Subsequently, we applied the inclusion and exclusion criteria to the candidate papers following the multi-stage selection process illustrated in Fig. 2. Initially, we selected papers based only on titles and abstracts, excluding those that met at least one of the exclusion criteria.

Papers that passed this stage were set aside for a further review, where the introduction and conclusion sections were carefully examined. If a paper provided sufficient evidence to meet the inclusion criteria I1 to I6, it was added to a list of included studies. After applying the selection criteria to the 1064 candidate papers, we selected 305 studies for quality assessment.

[2] https://www.capella-systems.com.

Table 3
Number of excluded studies.

SLR step	Selection criteria	Excluded
Study selection	Duplicated papers (E2)	2
	Short papers/conference proceedings (E4)	7
	Not related to RE (E5)	405
	Out of MBSE (E6)	82
	No processes for extracting or transforming presented (E7)	263
Quality assessment	QA score = 0 (No requirement in MBSE, w/ontologies or requirement extraction)	60
	QA score = 0.5 (Only requirement in the MBSE environment)	51
	QA score = 0.5 (Generating models from textual requirements but not extracting from MBSE)	9
	QA score = 1.0 (Only ontologies)	17
	QA score = 1.0 (Only requirement extraction in SysML)	37
	QA score = 1.0 (Generating models from textual requirement but not extracting from SysML)	14
Data extraction	No requirement extraction (E7)	13
	Secondary papers (E8)	7

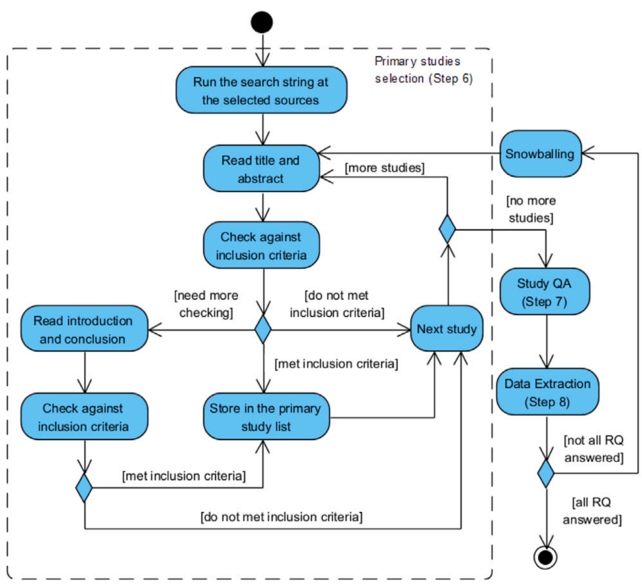

Fig. 2. Procedure for primary studies selection.

Table 4
Quality assessment checklist.

ID	Questions
Q1	Are the objectives clearly defined?
Q2	Does the work primarily focus on RE?
Q3	Are the methods clearly defined?
Q3a	If yes, can they be generalized to different conditions or stages?
Q3b	If yes, can they be integrated into different software?
Q3c	If yes, is this application empirically evaluated by specialists or in an industrial environment?
Q4	Does the present paper have a good literature review?
Q5	Is it a comparison with other presented works?
Q6	Are the requirements presented in an MBSE environment? (Yes = 0.5/No = 0)
Q6a	If yes, is the study supported by SysML (or Arcadia/ Capella)? (Yes = 0.5/No = 0)
Q7	Was an application developed specifically for the same purpose as this SLR? (Yes = 1.0/Partially = 0.5/No = 0)
Q8	Has an ontology in English been proposed to address the presented requirements issues in the extraction context? (Yes = 1.0/No = 0)

Note: The last four questions served as a threshold for including papers based on inclusion criterion I7.

Subsequent steps of our SLR resulted in additional exclusions. A summary of papers excluded at each stage of the SLR process is provided in Table 3.

3.3. Quality assessment

We conducted an evaluation of included studies with the intention of using only the evidence provided by high-quality studies. Following Kitchenham's guidelines (Kitchenham et al., 2009, 2015), we created a quality assessment checklist comprising eight main questions and four secondary questions, as presented in Table 4. The criteria were used as an overall indicator of the strength of evidence during data analysis.

We conducted team-based pilot assessment on a random sample of selected papers to mitigate potential biases from subjective assessment. The leading researcher used a think-aloud approach to explain their justifications to the assessment, while other researchers provided feedback and asked for clarifications until the quality criteria and assessment process were clear to all.

Following the recommendations of Kitchenham et al. (2015), we use a subset of the quality criteria (questions Q6 to Q8) as part of the inclusion criteria I7. Only studies that reached a minimum quality threshold of 1.5 out of 3 points were included (see Table 5). At the start of this quality assessment, we had 305 potentially included papers, however just 117 of them scored 1.5 or higher.

3.4. Data extraction

At the beginning of our data extraction process (Step 8 in Fig. 1), we identified that seven of the 117 included papers were secondary studies, and another 13 lacked sufficient information to address our RQs in terms of details on the requirement extraction process. As a result, we excluded these 20 papers based on exclusion criteria E7 and E8 (as shown in Table 3) and conducted the data extraction with 97 studies.

We used a data extraction form to support content analysis, systematically organizing the central questions of the studies, referred to as properties. Table 6 lists all properties and their corresponding research question(s). Properties P1 to P4 are not used to answer any particular RQ, but provide an overview of the included studies.

- **Approach description (P1)**: We extracted the full description associated with the objective of the study.
- **Solution presented (P2)**: We categorized the studies according to the results presented. Initially, we defined our strategy in three study categories: method, model, and prototyping tool. The studies that presented framework, method, technique, framework, process, or methodology were categorized as the *method* category. The *prototyping tool* category is appropriate for papers that build prototyping tools or applications that support software engineering practices. The *model* category refers to the studies presenting solutions as models focusing on transformation algorithms.

Table 5
Included studies and quality score.

Studies	Q6	Q6a	Q7	Q8	Score
Ballard et al. (2020), Bao et al. (2022)	Yes	Yes	Yes	Yes	3.0
Yang et al. (2021), Jahan et al. (2021), Iqbal and Bajwa (2016), Hamza and Hammad (2019), Tueno Fotso et al. (2018), Zhong et al. (2023)	Yes	Yes	Part.	Yes	2.5
Iqbal and Rehman (2019), Reis and da Silva (2018), Leopold et al. (2012)	Yes	No	Yes	Yes	2.5
Sanyal and Ghoshal (2018), Masuda et al. (2016), Elallaoui et al. (2018), Li et al. (2010), Yue and Ali (2012), Ibrahim and Ahmad (2010), Shimada et al. (2018), Sarkar et al. (2012), Rivero et al. (2019), Kornyshova and Deneckère (2010), Jyothilakshmi and Samuel (2012), Tseng and Chen (2008), Alkhader et al. (2006), Singh et al. (2009b), Lucassen et al. (2017), Sharma et al. (2015), Bajwa and Choudhary (2012), Ramackers et al. (2021), Osman et al. (2019), Arshad et al. (2019), Gupta et al. (2023), Abdelnabi et al. (2020), Maatuk and Abdelnabi (2021), Rago et al. (2016), Debnath et al. (2008), Pires et al. (2011), Ries et al. (2018), Bashir et al. (2021), Danenas et al. (2020), Sajjad and Sarwar (2016), Bajwa et al. (2012a), Elbendak et al. (2011), Li and Liu (2008), Kchaou et al. (2019), Krishnan and Samuel (2010), Patel and Priya (2014), Bajwa et al. (2007), Deeptimahanti and Sanyal (2011), Kumar and Sanyal (2008), Zhu et al. (2023), Tichy and Koerner (2010), Li et al. (2015a), Bryant and Lee (2003), Arif et al. (2022), Narawita and Vidanage (2016), Vasques et al. (2019), Jura et al. (2022), Naumchev et al. (2019)	Yes	No	Part.	Yes	2.0
Pröll et al. (2018), Sawant et al. (2014), Cardei et al. (2008)	Yes	Yes	No	Yes	2.0
Chen et al. (2022), Colombo et al. (2012), Brace and Ekman (2014), Klimek (2013), Colombo et al. (2011), Beimel and Kedmi-Shahar (2019), Carrillo et al. (2014), Bajaj et al. (2022), Wrycza and Marcinkowski (2011)	Yes	Yes	Part.	No	1.5
(Wei et al., 2016; Zhang et al., 2017; Debnath et al., 2011; Yue et al., 2010; Dahhane et al., 2014; Deeptimahanti and Babar, 2009; Gulia and Choudhury, 2016; Fatwanto and Boughton, 2008; Sharma et al., 2014; Yue et al., 2011a; Tahir and SarwarBajwa, 2013; Singh et al., 2009a; Ko et al., 2019; Harmain and Gaizauskas, 2000; Kaundal and Kaur, 2017; Njonko and El Abed, 2012; Zeni et al., 2015) Umber and Bajwa (2011), Šenkýř (2019), Bajwa et al. (2012b)	No	No	Part.	Yes	1.5
Effa Bella et al. (2019), Agt-Rickauer et al. (2019), Dong et al. (2012), Rago et al. (2009), Mehmood et al. (2019), Li et al. (2015b)	Yes	No	No	Yes	1.5

Table 6
Data extraction properties and related research questions.

ID	Properties	RQ
P1	Approach description	Overview of the studies
P2	Solution presented	Overview of the studies
P3	Research method	Overview of the studies
P4	Approach motivation	Overview of the studies
P5	Technology used to transform	RQ1
P6	RE phases	RQ2
P7	RE modeling styles	RQ2 & RQ3
P8	UML diagrams involved	RQ2 & RQ3
P9	Analysis models presented	RQ2 & RQ3
P10	Transformation approaches	RQ2 & RQ3
P11	Transformation results	RQ3
P12	Strengths	RQ5
P13	Weaknesses	RQ5
P14	Evaluation metrics	RQ5

- **Research method (P3)**: We categorized the studies according to the research method and their application context. Initially, we defined our strategy for this property based on the methodologies: case study, controlled experiment, quasi-experiment, and survey research, as defined by Easterbrook et al. (2008). The *case study* category refers to any observational study of a real-world case without performing an intervention. The categories of controlled experiment and quasi-experiment were unified into a single *experiment* category. That category is adopted if the study experiments according to Easterbrook et al. (2008). The *survey research* category encompassed studies that collect quantitative or qualitative data using a questionnaire. Besides that, we have defined two extra categories: illustrative scenario and not applicable. *Illustrative scenario* refers to the study that brings concise examples to evaluate their contributions. The *not applicable* is applied to studies discussing requirements extraction that did not use any research method. Regarding the application context, we defined this property in three categories: *industry*, *academic*, and *not applicable* or identified.
- **Approach motivation (P4)**: We extracted the complete motivation of the studies. Understanding the purpose of each paper can help identify opportunities, gaps, and areas for improvement in current and future engineering processes.
- **Technology used to transform (P5)**: This property is directly associated with RQ1, which aims to find tools, methods, techniques, or processes for extracting requirements from an MBSE environment. All technology presented explicitly throughout each study was extracted and stored.
- **RE phases (P6)**: The RE phases are relevant to identifying how the solution is applied in SE, answering RQ2. This property comprises five requirement activities described in the Software Engineering Body of Knowledge (SWEBoK) (Bourque et al., 1999) and Kotonya and Sommerville (Kotonya and Sommerville, 1998): elicitation, analysis, specification, validation, and management. *Elicitation* is the phase concerned with the origins of RE, in which their goal is understanding their sources and the writing techniques, being usually the first RE stage. *Analysis* is related to conceptual modeling, allocation, and solving conflicts between requirements. *Specification* refers to producing a document that can be systematically reviewed, evaluated, and approved. *Validation* is the phase in which requirement is subject to validation and verification procedures. Finally, *management* encompasses all the processes related to the RE life cycle.
- **RE modeling styles (P7)**: It refers to identifying the styles of software requirements modeling supported during the requirement transformation. Also, it is expected from the data collected to develop standards for requirement extraction.
- **UML diagrams involved (P8)**: The UML diagrams provide a standard notation to visualize the system's design. When transforming models, verifying the key elements and relationships among the models from the structure and behavior diagrams can be highly beneficial. This process can provide valuable insights and enhance understanding of the models' overall performance. By carefully analyzing the models' structure and behavior during extraction, you can gain a more profound knowledge of how they work and identify areas for improvement.

- **Analysis models presented (P9)**: This property aims to understand the kind of information or model constituted during the requirement extraction. The categories are divided into structure, behavior, or both. According to Yue et al. (2011b), *structure* emphasizes the system's static structure using classes, objects, attributes, operations, relationships, etc. In contrast, *behavior* emphasizes the system's dynamic behavior by showing interactions among objects, internal state changes, etc.
- **Transformation approaches (P10)**: This property aims to identify methods used in the extraction or transformation, which we categorized into rule-based, ontology-based, identity transformation, and pattern-based. Below are the definitions according to Yue et al. (2011b): *Rule-Based transformation* utilizes a set of predefined transformation rules. *Ontology-Based transformation* is built by processing NL sentences and acts as an intermediate model for ontology-based transformations. An *Identity Transformation* transforms a model into another model (target) without change in information content: the two models describe the same concepts but with different representations. *Pattern-Based* transforms source patterns into target patterns. A source pattern describes and organizes a set of source elements, while a target pattern describes and organizes a set of target elements.
- **Transformation results (P11)**: it refers to the automation's degree as the result of the transformation approach. For this property, we used four categories defined by Yue et al. (2011b): automated, automatable, semi-automated and manual. A transformation approach is *automated* if it has been fully implemented. When a transformation algorithm is proposed in a paper, we assess whether the description is detailed enough for it to be executed. If the description is sufficient, the transformation approach is considered *automatable*. However, in some cases, a transformation is *semi-automated* because user interventions are required. Finally, specific methods do not involve any form of automation, being categorized as a *manual* approach.
- **Strengths (P12)**: it refers to the state-of-art requirements extraction. We extracted data regarding each study's strengths and then examined the various benefits aids to develop a comprehensive understanding of the research's objective.
- **Weaknesses (P13)**: This property aims to extract the main risks, challenges, and improvements highlighted by the authors of each study. This process can assist in making an accurate and unbiased assessment of the research and its outcomes.
- **Evaluation metrics (P14)**: the objective of this property is to verify the methods and metrics used in the assessment of the result of each study. We extracted the descriptive analysis, variables, and statistical inference methods used.

Multiple researchers participated in this process in a collaborative way similar to the quality assessment. The main researcher used a think-aloud approach to extract data from a sample of papers, and the properties were discussed and refined as needed to ensure conformity across all studies.

3.5. Data synthesis

We used inductive coding to perform the data synthesis (Step 9 in Fig. 1), specifically for properties P4, P12, and P13. The inductive coding process is based on progressively emerging codes instead of starting from a set of predefined codes (Bailey and Jackson, 2003). Our coding process followed the two-cycle approach according to Matthew B and Michael (1994), as exemplified in Fig. 3.

The first step involved extracting or highlighting specific text related to a requested property in the data extraction form. Once the text was extracted, it was shortened into codes for more straightforward analysis.

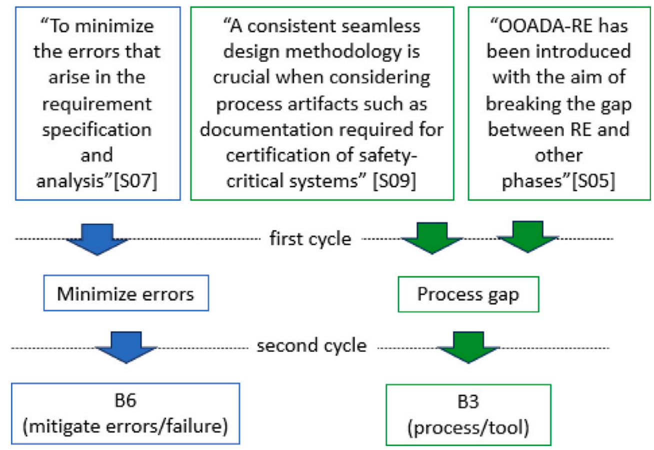

Fig. 3. Exemplar of the coding process for P4.

During a second step, the focus was on analyzing the emerging codes by identifying and merging similar codes to form categories. We repeated this process twice, and through it, we were able to create a comprehensive set of categories that accurately represented the extracted text and properties. This approach allowed for more efficient analysis and organization of the data, ultimately leading to more meaningful insights and actionable outcomes.

Regarding other properties, property P1 was used to extract the study's objective. The properties P5, P7, P8, and P14 were used to extract data and demanded a brief analysis to standardize the data. Finally, the properties P2, P3, P6, P9, P10, and P11 were pre-defined to ensure their suitability for the subsequent tasks.

3.6. Limitations and threats to validity

A common challenge for researchers is to guarantee the validity of the presented data. We have identified the main risks related to our study as follows:

- **Identification of primary studies**: We adopted a strategy to search a broad sample through electronic databases, as mentioned in Section 3.1. Initially, we collected a large number of potential studies, mainly from the Springer database, which compelled us to impose certain exclusion criteria to limit out dataset, such as considering only studies published in the English language and the software engineering category since 2010, to maintain the relevant results and the accuracy of the search query. In contrast, we choose inclusion criteria to filter irrelevant papers in the next step. Another important aspect to emphasize is the snowball approach, as shown in Section 3.1.3, that extended our search beyond limitations we found, leading to a more comprehensive review of the literature.
- **Validation of search strings:** The papers identified in our preliminary search were also found in the database search, which can be attributed to the formulation of the search string. Our approach to developing specific search strings was based on initial results from Google Scholar. We acknowledge that a more rigorous validation (such as the gold standard approach (Zhang et al., 2011; Kitchenham et al., 2015)) would better address potential biases and enhance the reliability of our database search.
- **Lack of more detailed description**: Another critical challenge we encountered is that some studies lacked detailed descriptions of the research approaches and solutions/technologies studied, which we noted during data extraction. This issue impacted the depth and comprehensiveness of our SLR. We attempted to mitigate this bias by applying a rigorous data extraction process, but we acknowledge that limitations on the evidence were beyond our control.

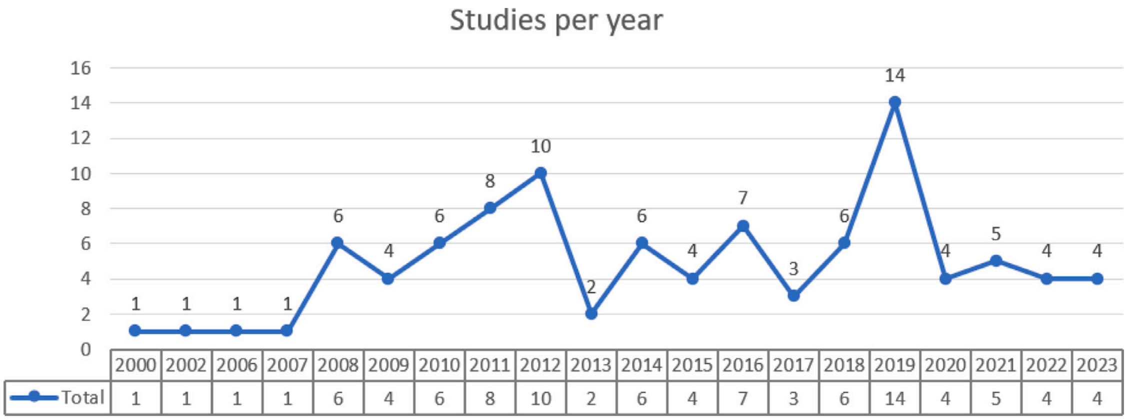

Fig. 4. Temporal view of the studies.

- **Data extraction consistency**: To manage the large amount of data generated during our research, we utilized a data extraction form as described in Section 3.4. The form was designed in a spreadsheet format and consisted of multiple columns, each of which pertained to specific properties referenced in the research. By filling out this form, we systematically organized and categorized the data for the synthesis data. We noted that the coding process in the synthesis data was susceptible to interpretation, which could lead to the generation of incorrect data. Therefore, coding was applied only on less relevant properties not related to core of our SLR. We utilized pre-defined options to ensure the adherence to core properties.
- **Publication bias**: Research has shown that there is a tendency to report positive outcomes more frequently than negative ones (Kitchenham et al., 2009). This trend can lead to biased conclusions, which can have severe implications in various fields of study. In our study, we observed a similar pattern, where more strengths than weaknesses were identified (see Section 4.5), highlighting a potential limitation of our findings. However, we made a conscious effort to report strengths and weaknesses side by side to transparently address any tendency toward this bias.

4. Results and analysis

As per the data synthesis, 97 studies have been included in this SLR, which discuss various tools, methods, techniques, processes, and models for requirements extraction or transformation in the context of SE.

Analyzing these 97 studies from a temporal point of view (Fig. 4), we can observe an increasing number of publications in the context of this review since 2006. The ongoing interest in model-based engineering can be attributed to the necessity and accessibility of advanced approaches for designing and developing complex systems aiming for practical models for analysis, simulation, and optimization.

P1. Approach Description: This property was adopted to assist in providing an overview of the researched article and in ensuring the consistency of data extracted from other properties.

P2. Solution Presented: The studies were categorized based on the P2 property, shown according to relevance order: Method (61.8%), Prototyping tool (36.1%), and Model (2.1%), as shown in Table 7. The first two attributes comprise almost all the studies and reflect directly our strategy searched.

P3. Research Method: In this property, we considered five categories (case study, experiment, illustrative scenario, survey, and not applicable) for the research method and three categories for application context (academic, industry, and not applicable). Fig. 5 shows the relationship between two properties: Solution presented (P2) and Research method (P3). Besides, it is possible to observe the relationship between the studies' research method and their application context in the same figure.

Regarding the presented data, we highlighted the use of case studies as a research method (48.4%), followed by experiment (25.8%), and illustrative scenario (18.5%). Concerning the application context, the vast majority of publications were employed in an academic setting (71.1%).

P4. Approach Motivation: Another aspect is to consider and comprehend the underlying motivation behind the studies to comprehensively evaluate the potential advantages that can be brought to the current process. By understanding the driving force behind the research, we can gain a deeper insight into the intended outcomes and how they can be effectively applied to the current process. We categorized the purposes into six relevant themes, as explained in the P4 property and shown in Fig. 6.

In some studies, more than one motivation was found, or secondary purposes were presented; the data were accounted for in both cases. Despite this disclaimer, it is worth noting that category B5, represented by productivity and result, was well highlighted between the address themes with 50 citations. *Time consumption* (19), *reducing effort* (14), *saving money* (6), and *automated support* (6) were the most cited among the subcategories.

Next in frequency, there are two categories: quality and accuracy (B1) and knowledge and technology (B4), each with 25 citations. The first category (B1) has the most cited subcategories, *accuracy*, and *consistency*, with 14 and 7 appearances, respectively. About the B4 category, *solving complexity* (14) and *missing expertise* (8) were the most cited subcategories.

For the B3 category, there are 23 studies, with the most frequent subcategories: *process gap* and *no tools available*, with 13 and 5 appearances, respectively. The B2 category, addressed by completeness and reliability, received 20 citations, with the most cited subcategories being *completeness* (8 citations), *missing requirement*, and *solving ambiguity* (both with four citations).

Finally, the B6 category is characterized by the category mitigating errors and failure and contains 17 citations. Specifically, it is associated with preventing project failure (7 appearances), mitigating errors (4 appearances), and correctness (4 appearances).

After analyzing the motivation behind the studies cited above, it becomes clear that productivity is one of the most widely discussed themes on a global scale. With the advancement of technology and the rise of competition, organizations worldwide seek ways to enhance their productivity levels while reducing development time. This has led to a renewed focus on efficient processes, optimized workflows, and innovative solutions, contributing to the impressive search results focused on this topic.

Table 7
Solution presented (P2).

P2	Studies	Count	%
Method	Wei et al. (2016), Zhang et al. (2017), Debnath et al. (2011), Dahhane et al. (2014), Gulia and Choudhury (2016), Fatwanto and Boughton (2008), Pröll et al. (2018), Effa Bella et al. (2019), Sharma et al. (2014), Sanyal and Ghoshal (2018), Elallaoui et al. (2018), Ballard et al. (2020), Colombo et al. (2012), Dong et al. (2012), Kaundal and Kaur (2017), Shimada et al. (2018), Brace and Ekman (2014), Rivero et al. (2019), Kornyshova and Deneckère (2010), Jyothilakshmi and Samuel (2012), Rago et al. (2009), Mehmood et al. (2019), Iqbal and Rehman (2019), Alkhader et al. (2006), Lucassen et al. (2017), Klimek (2013), Njonko and El Abed (2012), Sharma et al. (2015), Bajwa and Choudhary (2012), Li et al. (2015b), Osman et al. (2019), Colombo et al. (2011), Leopold et al. (2012), Jahan et al. (2021), Iqbal and Bajwa (2016), Abdelnabi et al. (2020), Maatuk and Abdelnabi (2021), Hamza and Hammad (2019), Debnath et al. (2008), Beimel and Kedmi-Shahar (2019), Carrillo et al. (2014), Pires et al. (2011), Umber and Bajwa (2011), Cardei et al. (2008), Tueno Fotso et al. (2018), Danenas et al. (2020), Sajjad and Sarwar (2016), Kchaou et al. (2019), Krishnan and Samuel (2010), Deeptimahanti and Sanyal (2011), Šenkýř (2019), Wrycza and Marcinkowski (2011), Zhu et al. (2023), Tichy and Koerner (2010), Bajwa et al. (2012b), Arif et al. (2022), Narawita and Vidanage (2016), Vasques et al. (2019), Jura et al. (2022), Naumchev et al. (2019)	60	61.8
Prototyping tool	Yue et al. (2010), Deeptimahanti and Babar (2009), Agt-Rickauer et al. (2019), Yue et al. (2011a), Tahir and SarwarBajwa (2013), Singh et al. (2009a), Chen et al. (2022), Masuda et al. (2016), Ko et al. (2019), Li et al. (2010), Yue and Ali (2012), Ibrahim and Ahmad (2010), Harmain and Gaizauskas (2000), Sarkar et al. (2012), Sawant et al. (2014), Reis and da Silva (2018), Yang et al. (2021), Singh et al. (2009b), Ramackers et al. (2021), Zeni et al. (2015), Arshad et al. (2019), Gupta et al. (2023), Rago et al. (2016), Ries et al. (2018), Bashir et al. (2021), Bajaj et al. (2022), Zhong et al. (2023), Bajwa et al. (2012a), Elbendak et al. (2011), Li and Liu (2008), Patel and Priya (2014), Bao et al. (2022), Kumar and Sanyal (2008), Li et al. (2015a), Bryant and Lee (2003)	35	36.1
Model	Tseng and Chen (2008), Bajwa et al. (2007)	2	2.1

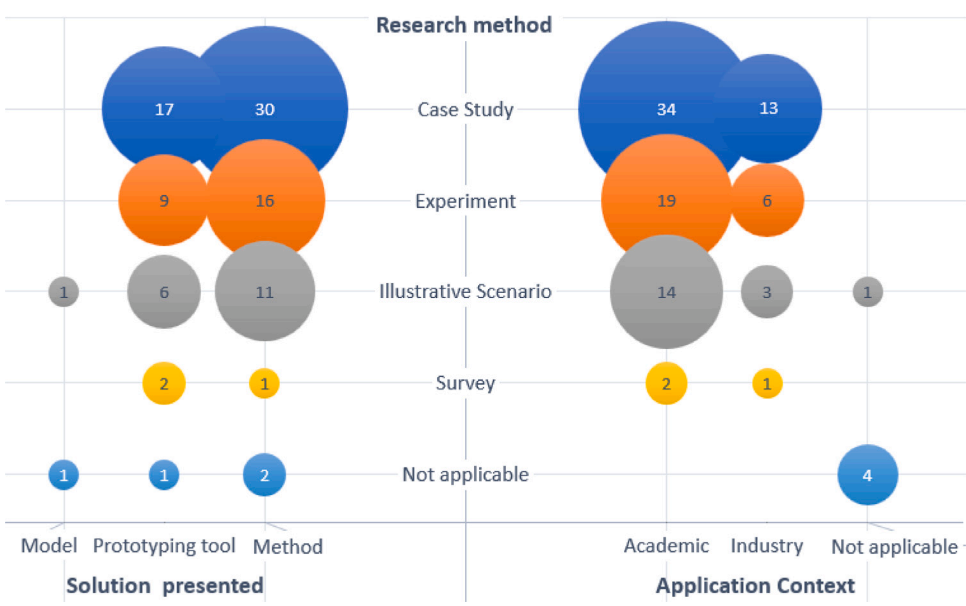

Fig. 5. Research method.

4.1. Tools, methods, techniques or process for requirements extraction (RQ1)

P5. Technology Used to Transform: Upon completing the data extraction, we conducted a thorough analysis of the various technologies in use. We classified these technologies based on their similarities in terms of attributes, resulting in the creation of the following categories: processes, languages, tools, metamodels, and databases.

The process category includes NLP techniques, tasks, or applications relevant to this topic. *Languages* refer to programming or ontological languages used directly for extraction tasks. We defined *Tools* as all application programs developed to perform specific tasks other than extraction, typically managed by end-users. In the *Metamodels* category, we include any models or frameworks that define a modeling language in which a model can be expressed. Finally, we considered any organized data collection, such as glossaries, as a *Database*. This categorization helped us to comprehensively organize the various technologies used within P5 and ensure a detailed stratification according to Table 8. It is worth noting that eleven out of the 97 included studies did not mention any specific tool, method, technique, or process for requirement transformation.

Among the reported databases, we discovered that Wordnet was the most frequently cited, appearing 13 times. It was the only database mentioned repeatedly, indicating its importance and relevance. WordNet (Miller, 1995) is a lexical database of English in which nouns, verbs, adjectives, and adverbs are grouped into sets of cognitive synonyms, each expressing a distinct concept. These cognitive synonyms are interlinked using conceptual-semantic and lexical relations. WordNet's structure makes it a helpful tool for NLP. Wordnet has been employed as a semantic dictionary for correctness validation (Abdelnabi et al., 2021) or as a word sense disambiguation determined by the sense of each word in a text (Sànchez-Ferreres et al., 2017).

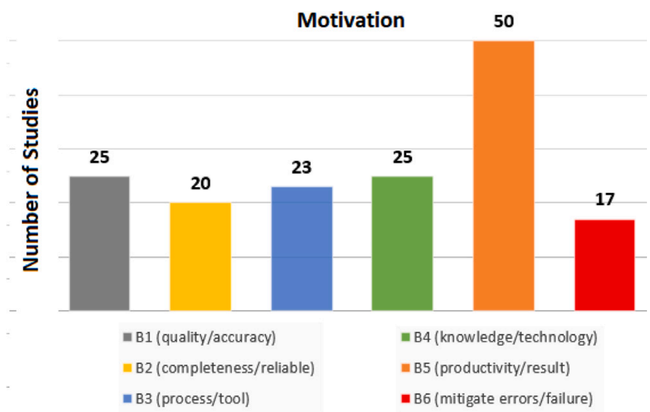

Fig. 6. Motivation of the studies per category.

Table 8
Technologies used for requirement transformation.

Technology		Count
Database	Wordnet	13
Language	OWL/OWL2	8
	Java	2
	Prolog	2
	Python	2
Metamodel	SBVR	9
Process	POS tagging	33
	Tokenization	20
	Sentence splitting	10
	Text chunking	6
	Lemmatization	6
	NER	2
	Stop word removal	2
Tool	Stanford Parser	18
	Stanford POS Tagger	6
	OpenNLP	5
	Eclipse IDE	4
	aToucan	3
	NLTK	3
	Protégé	3
	Stanford CoreNLP	3
	JavaRAP	2
	OpenIE	2

Note: We listed only the technologies that occurred more than once.

We found Web Ontology Language (OWL) among the most cited Languages with eight instances. OWL is an ontology-related language, one of the technologies defined by the World Wide Web Consortium (W3C) for the Semantic Web (McGuinness et al., 2004). OWL is a family of knowledge representations used for specifying ontologies.

Other *Languages* related to our research are well-known programming languages, such as Java and Python, with two use cases. Prolog, a logic programming language focused on the original intended field of use, NLP, was also mentioned twice. One of the possibilities for the low number of registrations of programming languages is that these languages are incorporated within the tools or software, which are also reported as a product of these applications, or even have contributed in implicit ways such as an Application Programming Interface (API). Another possibility we considered for this omission is due to the significance of the topic at hand.

Regarding Metamodels, the Semantics of Business Vocabulary and Business Rules (SBVR) (OMG, 2019) was mentioned in 9 instances. SBVR is an adopted standard from the OMG that defines a metamodel for developing semantic models of business vocabularies and business rules using NL (Njonko and El Abed, 2012). SBVR provides a way to capture specifications in NL and represent them in formal logic, enabling machine-processed based on a combination of linguistics, logic, and computer science (OMG, 2019). Similar to the database category, SBVR was the only metamodel referenced more than once.

About Processes, Part-of-speech (POS) tagging was the most frequently mentioned process, with 33 instances. Tokenization was mentioned 20 times, followed by sentence splitting (10 instances), text chunking (6 instances), and lemmatization (6 instances). The list was completed with Named Entity Recognition (NER) and stop word removal, both mentioned twice. Here is a brief definition of the most referenced processes:

- POS tagging is an NLP task related to morphological analysis that involves marking a word in a text corresponding to a particular part of speech based on its definition and context. This task is complex because some words can represent different parts of speech depending on their usage, and some parts of speech can have multiple meanings. The high number of instances cited for POS tagging directly results from its primary purpose: to extract meaningful information from text data and enable various requirement transformations.
- Tokenization or word segmentation, on the other hand, is an NLP task related to text and speech processing. The process consists of separating a chunk of continuous text into separate words and separating them with spaces.
- Sentence splitting is an NLP task related to syntactic analysis. The process consists of dividing the text into sentences and finding the sentence boundaries, which can be done using rules or models.
- Text chunking extracts meaningful groups of words based on their syntactic structure. Chunking typically follows tokenization in the NLP pipeline and is often used as a preprocessing step for tasks such as named entity recognition, parsing, or information extraction.
- Lemmatization is another NLP technique related to morphological analysis that involves reducing words to their normalized form.

Regarding the *Tool* category, the Stanford Parser has been mentioned prominently by 18 studies. Follow this are: Stanford POS Tagger (6), OpenNLP (5), Eclipse Integrated Development Environment (Eclipse IDE) (4), aToucan, Natural Language Tool Kit (NLTK), Protégé, and Stanford CoreNLP, each with three instances. The Stanford NLP Group developed some NLP software cited, such as Stanford Parser, Stanford POS Tagger, and Stanford CoreNLP. Concerning other tools:

- OpenNLP is an open-source NLP Java library, toolkit-based for supporting NLP tasks;
- Eclipse IDE is widely regarded as the Java development environment;
- NLTK is a Python library for symbolic and statistical NLP;
- Protégé is an open-source ontology editor and a knowledge management system; and
- aToucan (Yue et al., 2015) is an automated framework that generates a UML analysis model from use case models.

Except for aToucan, which is tool driven for extracting requirements, we noticed that other tools mentioned are primarily focused on various NLP tasks or act as a development platforms in the case of Java programming language. One point that we highlighted is that these tools are primarily used for transforming unstructured text to the model direction, and rules-based methods mainly drive the process of extraction or transformation.

4.2. Approaches for requirements extraction applied in SE (RQ2)

In Section 4.1, we identified the tools, methods, processes, and techniques used for requirements extraction, which answers RQ1. RQ2 complements the first question by identifying the application of these approaches in SE. We structured the searches into three pillars during the requirement extraction: identify the RE phases that are applied (i), the RE modeling styles that are used (ii), and the kinds of transformations applied (iii).

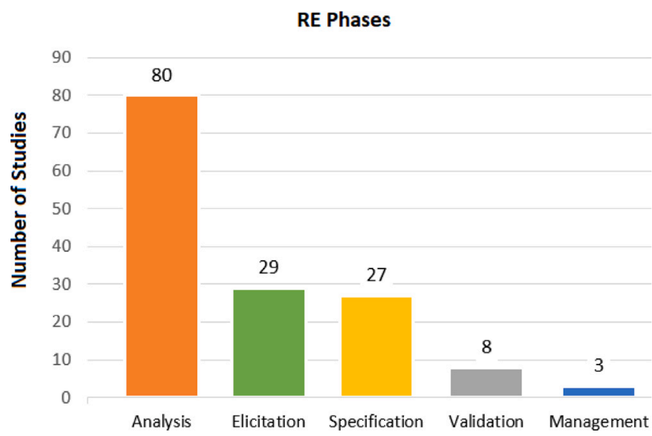

Fig. 7. RE phases.

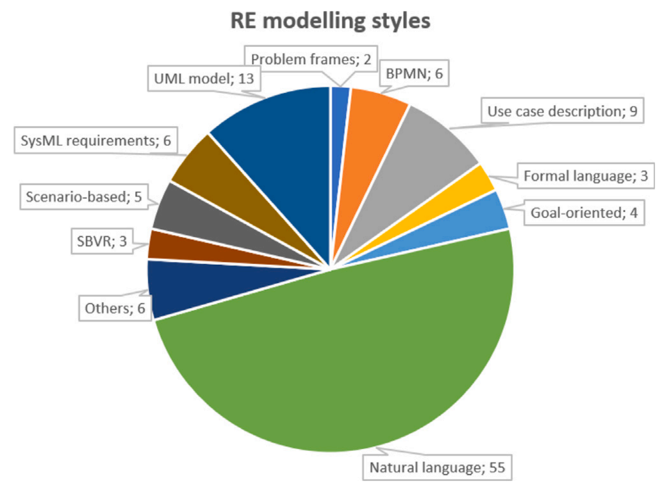

Fig. 8. RE modeling styles.

Fig. 7 shows the RE phases found in the selected studies. In certain studies, their methods were implemented in various or even all RE phases, and we considered all of these occurrences on the graph mentioned above. We expected a majority number to be assigned to the analysis and specification phases; however, the predominance was strictly to the analysis phase with 80 instances of 97 studies. Out of the 17 studies not included in the analysis phase, one (Mehmood et al., 2019) did not specify which phase it belongs to. One study covered the validation phase, one covered the management phase, six covered the specification phase, and eight covered the elicitation phase.

It is worth observing the high number of studies performed in the elicitation phase (29 instances), demonstrating the usefulness of these technologies applied at the beginning of the RE. Multiple methods outlined in this SLR can clarify this value. For instance, these approaches can transform unstructured text into a requirements model during the elicitation phase.

Another point to highlight is the low number of studies related to the management phase. We consider this number to be due to the need for processes such as model extraction to be incorporated into the main RE management tools. As such processes are consolidated with the advent of shared solutions, these solutions are expected to be made available and embedded in these tools soon.

P7. Regarding the RE modeling styles We identified the standard RE modeling used during requirements transformation through extracted data from the P7 property. We aim to comprehend which styles are generally supported in extraction and which are the most attractive for the SE community.

We perform the data synthesis for the P7 property in two steps. The first step was identifying the elements that preceded and followed the transformation. Next was to determine whether both were characterized as requirements. In the positive case, we extracted both styles in our analysis. Otherwise, if only one of the elements was considered a requirement, we extracted only that one style. Fig. 8 displays the different RE modeling styles identified during the requirements extraction process.

Most requirements are presented in NL (49%). The need to reduce ambiguity and ensure accuracy in statements written in NL explains the significant number of NL requirements used as a primary source of that transformation.

Other RE modeling styles are highlighted in this search, but only partially as a primary source. UML models were the second most used RE modeling style (13 instances, 12%), followed by use case descriptions (9 instances, 8%).

Although SysML requirements are also a model, we did not consider them a UML model in this property (P7). The main reason is that requirement diagrams are specific to SysML; thus, we intended to compare UML models and SysML represented exclusively by requirements diagrams. The SysML requirement style resulted in six papers (5%).

We mapped to determine whether the RE modeling style is a source or a target for the observed extractions. We thoroughly traced every relationship between the RE modeling styles depicted in Fig. 9 during the mapping process. In this figure, each RE modeling style on the left is identified as the source of transformation, with the RE modeling style on the right as the target. The number inside the circle represents the total frequency of that style in the studies. The identification number next to the arrow, located on the arrow lines at the top, signifies the frequency of that transformation between the identified styles. The absence of a number next to the arrow indicates a frequency of one.

Concerning the data, the UML model and NL are highlights between the RE modeling styles presented, mainly whether we observe the high number of transformations from NL to UML (35.6%). The category labeled *Not translated*, represented as a RE modeling style target in Fig. 9, refers to studies focused on refining, tracing, improving, and analyzing models.

P8. UML Diagrams & P9. Analysis Models: Exploiting this comparison, we established the P8 property, which aims to find requirement extraction from MBSE, as described in Section 3.4. Unlike the P7 property, the UML and SysML were considered for this property because both have similar functions from the perspective of MBSE.

A list of the diagrams we found is presented in Fig. 10. The direction of the transformation identify the UML/SysML diagram's exact role in the transformation process. Either it functions as a primary source (from the UML/SysML diagram) or as a product of the transformation (to the UML/SysML diagram).

After analyzing the UML/SysML diagrams involved, we have highlighted the predominance of the UML class diagram, with 44 mentions, representing 37,3% of the collected data. However, concerning that highlight, most transformations were recorded towards the UML class diagram as a resulting product (32 mentions). The transformation to this kind of diagram reflects a considerable need in the SE community, mainly when it arises from unstructured statements.

The second diagram most cited was the UML use case diagram (ucd), with 31 instances. We observed a slight preference for the use case diagram as a resulting product (to the UML use case diagram) with 18 cases compared to 13 cases from the UML use case diagram.

Of all the diagrams, the UML activity diagram (act) was the third most cited (13 instances). The preference by the resulting product (to the UML activity diagram) is evident instead of the opposite direction (10 cases against three, respectively).

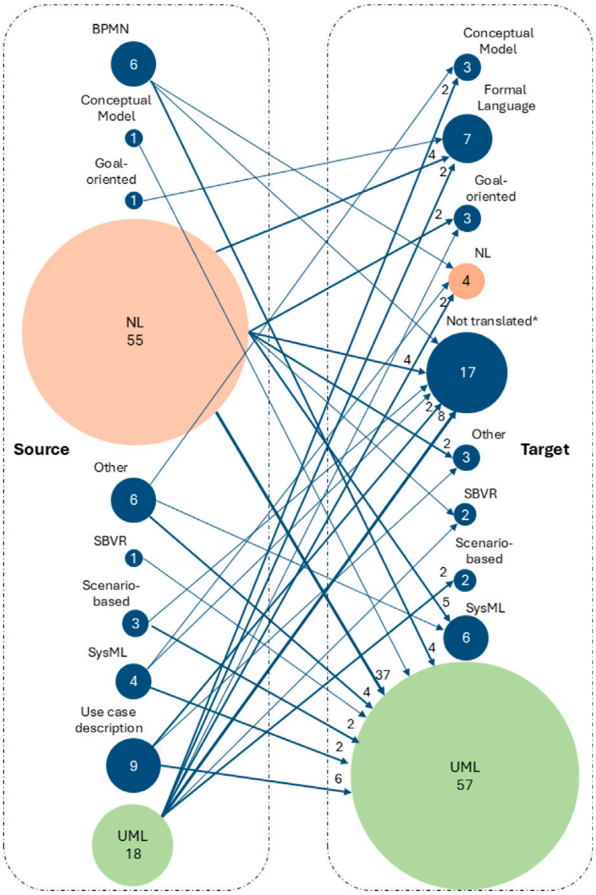

Fig. 9. RE transformation map.

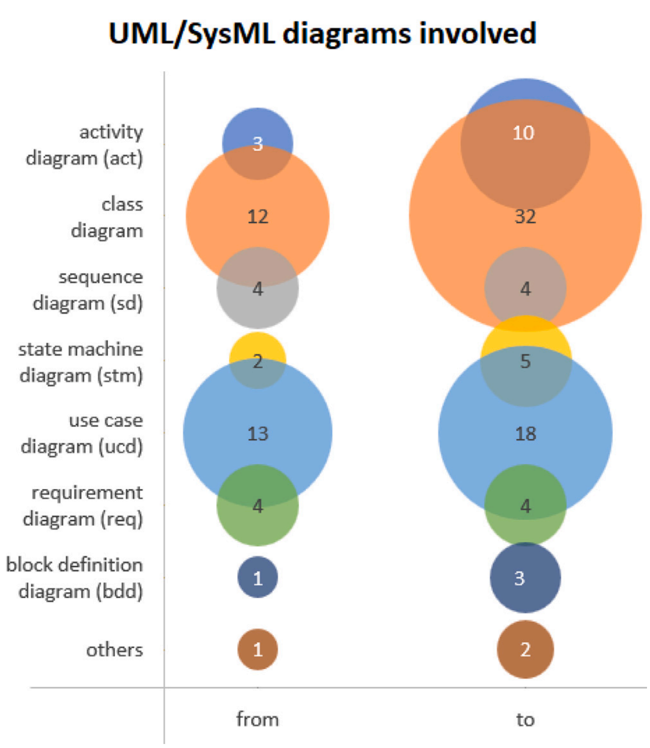

Fig. 10. UML/SysML diagrams involved.

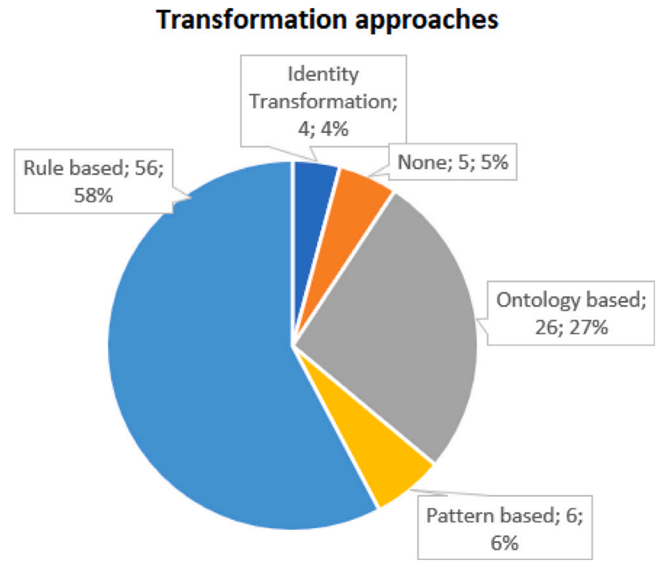

Fig. 11. Transformation approaches.

P10. Transformation Approach: For the third and final pillar of RQ2, we examined the techniques used in the extraction or transformation, categorized according to the P10 property (Section 3.4). Fig. 11 illustrates the distribution of identified transformation approaches in the selected studies.

The most commonly used transformation approach is rule-based, accounting for 58% of the studies (56 instances). The ontology-based transformation comes in second with 27% (26 cases), while others account for 10% (10 cases). The transformation approach was not found in 5 studies.

Out of the four transformation approaches presented, we highlighted the rule-based approach, characterized by the need to structure the statements during transformation. It has also often been used to support the transformation into a UML model, generating from the text to the model.

4.3. Approaches for requirements extraction applied to support textual requirement (RQ3)

In the previous subsection, we addressed three pillars that aim to identify the approaches in SE. To answer RQ3, we went into more detail on approaches that support textual requirements. To accomplish this, we applied the identical principles used in RQ2, which were focused on NL. We filtered out RE modeling styles, specifically selecting studies related to NL.

Firstly, we analyzed the UML/SysML diagrams used during extraction and compared them with the requirements represented in NL. We also considered the direction of this transformation to clarify how these approaches are applied to support textual requirements. Fig. 12, through the bubble chart, represents the comparison above.

It is noteworthy a predominancy of studies using the UML/SysML diagrams as a target (53 instances) compared with the UML/SysML diagrams used as a primary source (9 instances). This discrepancy can be explained by the need to reduce or avoid ambiguous statements. NL processing requires extra effort to minimize ambiguity and is an error-prone task.

The UML class diagram is the most cited under this condition, with 27 instances. The proportion of UML class diagrams used as a product and primary source is precisely eight times, with 24 and 3 instances, respectively. This proportion is even more prominent than that of all RE modeling styles in the same diagram, which is 32/12, according to Fig. 10.

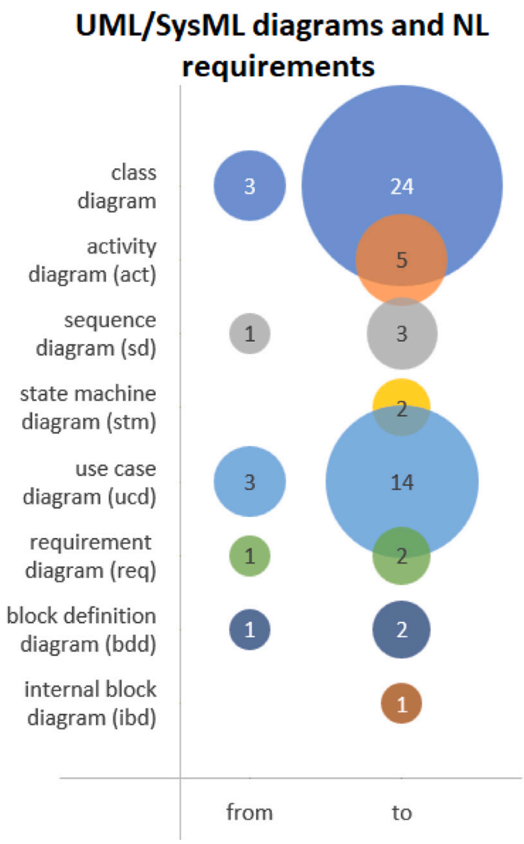

Fig. 12. UML/SysML diagrams and NL requirements.

Next in frequency, the second diagram most cited was the UML use case diagram, with 17 citations. The use of this diagram remains relevant in the particular transformation with NL in comparison to all RE modeling styles. However, the proportion has increased, with 14/3 analyzing only NL and 18/13 all RE modeling styles.

We noticed that there should be a significant difference between the structure and behavior diagrams, but neither indicates a preference in the SE community related to NL.

In the next pillar, we examined the transformation approaches used during extraction involving UML/SysML diagrams and compared them with requirements represented in NL. Following the early analysis, we observed the direction of the transformation. However, instead of the previous analysis presented, we considered two options for the transforming direction concerning the NL requirement: from NL or to NL. Fig. 13 shows a bubble chart representing the comparison mentioned.

Out of 49 transformation approaches identified, 40 used an NL as a primary source, representing 81.6%. NL was used as a transformation product in the remaining nine instances. We also noted that most transformation approaches cited refer to rule-based approaches, with 30 cases (26 from NL and 4 to NL). The above data show that rule-based approaches are essential to generate models from NL requirements. On the other hand, the low number of citations prevents us from drawing conclusions in the opposite direction.

P11. Transformation Results: This property aims to verify the level of automation achieved through the transformation methods mentioned earlier. We have gathered data on the transformation results from UML/SysML diagrams to NL requirements or vice-versa. The comparison between these results is depicted in Fig. 14.

We have observed that the vast majority of transformation results, 83% of them, are executed automatically, with a total of 39 cases. This number is in line with best practices worldwide. Automating processes can improve efficiency and accuracy and save costs while allowing humans to focus on tasks that add more value.

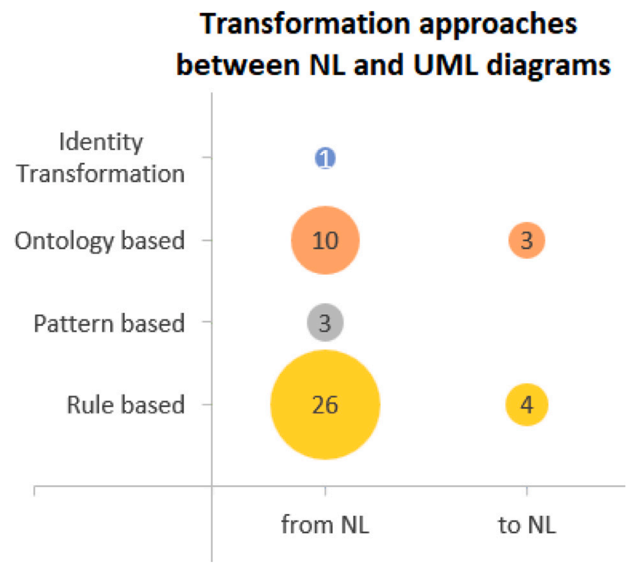

Fig. 13. Transformation approaches between NL and UML diagrams.

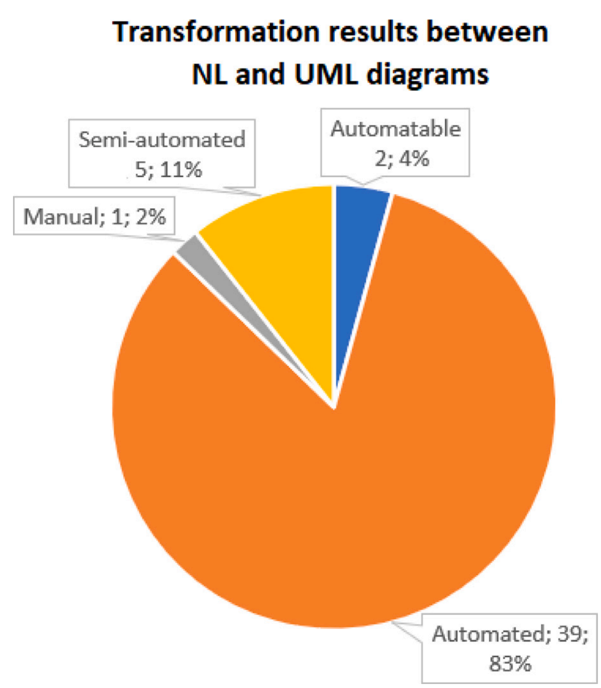

Fig. 14. Transformation results between NL and UML diagrams.

4.4. *Approaches for requirements extraction applied in the Arcadia method (RQ4)*

As mentioned in Section 3.1.3, aside from the search strategy allocated for this SLR, we generated a forward snowballing in search of the studies that could bring approaches addressed to extraction requirements in the Arcadia Method. However, we have yet to find papers related to this theme. The most covered topics concerning the Arcadia method refer to using Arcadia/Capella for systems modeling, use in the verification and validation phases, and other engineering issues.

We assumed that if there are papers discussing modeling in the Arcadia method but not related to extraction requirements, it could indicate that the modeling is in the implementation or exploratory phase. The timeline following the launch of the Arcadia Capella and the

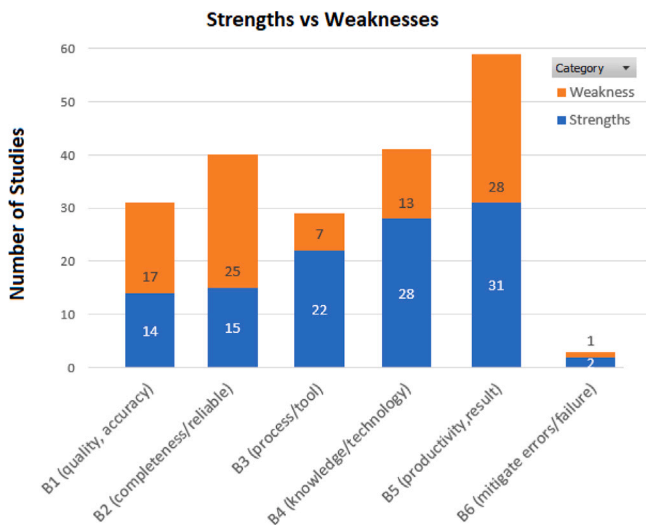

Fig. 15. Strengths & Weaknesses per category.

rising number of publications on this topic indicate that the approach is advancing towards relevant aspects of RE modeling. Automating requirements often arise from extensive usage and engagement with a system or process.

4.5. Benefits and difficulties about requirement extraction (RQ5)

Following the previous section, every strength and weakness explicitly mentioned by the authors in the content of each study was attributed to the P12 and P13 properties. The data extraction process was based on inductive coding, as cited in Section 3.4. Then, the data were separated using the same categories adopted for the P4 property (motivation of solution) to establish a comparative analysis between the benefits and needs of improvements.

P12. Strengths & P13. Weaknesses: Out of the 97 studies analyzed for data extraction, we discovered that four studies did not mention any strengths. In contrast, 12 studies did not mention any weaknesses or scopes for future improvement.

Fig. 15 compares the earlier-mentioned properties in the context of requirements extraction. The B5 category represented by productivity and results has more occurrences (31 strengths and 28 weaknesses, 59 in total), behaving the same way that category referred to motivation, occupying the first position among these properties.

The most commonly mentioned strengths in the B5 category are *saving time* (8), *saving effort* (5), *quick generation* (3), and *efficiency* (3). On the other hand, the most common weaknesses mentioned are *no extraction at all* (12) and *incomplete extraction* (6).

The second most frequently cited category was the B4 category, which pertains to knowledge and technology, with 41 appearances (28 strengths and 13 weaknesses). The most commonly cited strengths in this category were *better understanding for analysts* (5) and a *good demonstration of the NLP process* (3). Nevertheless, the most frequently mentioned weakness was *human dependency* (3).

The third most cited category was B2, defined by completeness and reliability, appearing 40 times. It comprises 15 strengths and 25 weaknesses. Among the strengths, *traceability* (5), *finding ambiguous or missing requirements* (3), and *completeness* (3) were the most frequently cited. However, the most commonly reported weaknesses were the *limitations for complex statements* (9) and the *need for more transformation rules* (6).

The following category (B1) was represented by quality/accuracy reported 31 times (14 strengths and 17 weaknesses). The strengths were mainly related to *consistency* (4) and *accuracy* (3), while the weaknesses mainly were related to *missing evaluation data* (8) and *limited accuracy* (4).

The last two categories were B3, represented by process/tool, which had 29 mentions (22 strengths and seven weaknesses), and the B6 category, represented by mitigate errors/failure, which had just three mentions (two strengths and one weakness). There were no outstanding features in the subcategories of either group.

Analyzing the presented data, we noted that the B5 category referred by productivity/result has a significant mention associated with the real purpose of the studies. In recent times, the pursuit of efficiency and optimization in various processes has been largely employed by the practices of major corporations across the globe. These companies have been at the forefront of driving advancements in this direction, inspiring others to follow suit and adopt similar strategies to save time and cost and reduce efforts in their activities.

One relevant point to consider is the inequality in the B4 category, which is mentioned frequently as a strength but rarely cited as a weakness. After reviewing the studies' results, it is increasingly difficult to find negative reviews about the knowledge or technology used.

It is essential to highlight the high number of areas for improvement related to the completeness and reliability of the tools/methods used and the low number of studies that aim to quality, accurate, or mitigate failure.

P14. Evaluation Metrics: Regarding the RQ5, we verified the methods and metrics used in assessing the result adopted by each study. We extracted the data through the P14 property and listed it in Table 9.

Table 9
The ten most cited evaluation methods and metrics in requirements transformation.

Evaluation Metric	Frequency
Precision	29
Recall	28
F-measure	17
Accuracy	6
Coverage	4
Survey	4
Technical comparative	3
Descriptive statistic	3
Over-specification	3
Semantic evaluation	3

We noted that precision (29) and recall (28) are the most cited methods for evaluation. In information retrieval, precision is the fraction of relevant instances among the retrieved instances. At the same time, recall is the fraction of relevant instances retrieved over the total amount of relevant instances. Both precision and recall are based on an understanding and measure of relevance defined by Allen et al. (1955). The two measures are combined in the f-measure (third most cited, with 17 instances) to provide a single measurement for a system.

5. Conclusion

This paper presented an SLR that examines approaches in the literature for extracting requirements from an MBSE environment or model generation from NL requirements. The review uncovered several key findings that have important implications for future research, as follows:

Dominance of the model generation from NL. Regarding the RE modeling styles searched in this SLR, the vast majority, 54 instances, refer to NL. Given that productivity is a primary goal of these studies, this high number indicates the importance of managing NL requirements as input or output. When comparing the UML/SysML diagrams used for extracting requirements associated with NL requirements, we observe a preference for utilizing the NL as input for transformations. This scenario highlighted class and use case diagrams among the observed population with 38 appearances. When analyzing the transformation results, we must consider the numerous automated approaches identified in 39 instances.

Another point to discuss is the relevant use of rule-based transformation approaches, considering only the transformations that involved NL requirements with 26 instances. In several cases, the rules for extracting data depend on the NLP techniques. These techniques directly relate to the technologies, whether embedded into a program or code or used within a methodology.

Regarding the RE phases observed, it is relevant to the number of studies performed in the elicitation phase (29 instances), demonstrating the usefulness of these technologies applied at the beginning of the RE phase. However, this number could be higher given that this review focused on extracting from MBSE requirements, supported by exclusion criteria E5, which removes studies unrelated to RE between analysis and specification phases.

Transforming model-based requirements to NL requires more data. We have gathered a significant amount related to extracting requirements from NL; however, we have yet to collect as much content regarding the opposite direction. When we compare the two transforming directions, the number of studies for transforming model-based requirements to NL is considerably lower than the number for the opposite direction, with a proportion of 1:6.

The limited amount of data for requirements extraction from MBSE hinders our ability to correlate it with attributes such as transformation approaches or transformation results. Similarly, we could not correlate this extraction with any UML/SysML diagrams. One possible reason for this disparity could be related to automating RE elicitation. The effort is justified by the significant number of UML/SysML diagrams generated from NL due to the need to examine several documents and statements.

Requirement extraction in Arcadia/Capella needs more evidence. Despite conducting a comprehensive search, we have yet to find papers on extraction requirements in the Arcadia/Capella. We used to find papers mainly focused on using Arcadia/Capella for systems modeling, verification and validation, and other engineering issues. The lack of papers on extraction requirements suggests that modeling in the Arcadia/Capella may be in the implementation phase.

Because Arcadia/Capella has shown many capabilities and resources for the development phase in SE, we expect that it will soon progress to automating requirements in the model-based environment. Furthermore, we also expected more validation from industrial practitioners, with attributes offering more substantial evidence rather than relying on use case studies.

State-of-art and state-of practice: Based on our analysis, most of the studies were conducted in academic settings, which suggests that the findings may not reflect the state-of-practice in real-world applications. This discussion is crucial for understanding the applicability of the research findings. Future studies should aim to include more in-vivo research to bridge this gap and provide a more comprehensive understanding of the current state-of-practice.

Research agenda. The results presented in this SLR can benefit the SE community since it gathers and summarizes evidence from the primary studies included in the review, forming a body of knowledge regarding the requirements extraction from MBSE. We propose a research agenda to the SE community based on the sequence of this SLR. Some issues that need to be considered include:

- Search for detailed transformation results: The transformation results presented in this SLR had specific characteristics but did not delve into how automation was carried out. On the other hand, the summarized studies can provide an initial discussion about this suggested theme.
- Collect more studies focused on requirements extraction from MBSE: The number of studies included in this SLR needed to be increased for statistical analysis. With more studies, it may be possible to draw inferences and establish correlations related to the mentioned topic.
- Examine upcoming Arcadia/Capella updates and features: This tool brings significant advances to modeling systems, many of them expressed from journals and magazines of interest to the SE community; however, until the moment of this SLR publication, without any presented requirements extraction theme. Due to being a collaborative tool, we believe that Arcadia/Capella will transition to automation requirements in the MBSE environment shortly.

CRediT authorship contribution statement

Jefferson L. Santos: Writing – original draft, Visualization, Investigation. **Luiz Eduardo G. Martins:** Validation, Supervision, Methodology, Conceptualization. **Jefferson Seide Molléri:** Writing – review & editing, Supervision, Methodology, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. During the preparation of this work the authors used Grammarly in order to improve readability. After using this tool, the authors reviewed and edited the content as needed and took full responsibility for the content of the publication.

Data availability

The SLR dataset is available at www.zenodo.org/.

References

Abdelnabi, E.A., Maatuk, A.M., Hagal, M., 2021. Generating UML class diagram from natural language requirements: A survey of approaches and techniques. In: 2021 IEEE 1st International Maghreb Meeting of the Conference on Sciences and Techniques of Automatic Control and Computer Engineering MI-STA, Tripoli, Libya.

Ahmed, A.S., Ahmed, A., Eisty, N.U., 2022. Automatic transformation of natural to unified modeling language: A systematic review. In: 2022 IEEE/ACIS 20th International Conference on Software Engineering Research, Management and Applications. SERA, Las Vegas, NV, USA.

Allen, K., Berry, M.M., Luehrs, Jr., F.U., Perry, J.W., 1955. Machine literature searching VIII. Operational criteria for designing information retrieval systems. p. 93, American Documentation (pre-1986), 6 (2).

Bailey, D.M., Jackson, J.M., 2003. Qualitative data analysis: Challenges and dilemmas related to theory and method. Am. J. Occup. Ther. 57 (1), 57–65.

Biolchini, J., Mian, P.G., Natali, A.C.C., Travassos, G.H., 2005. Systematic Review in Software Engineering. Technical Report ES, vol. 679, no. 05, System Engineering and Computer Science Department COPPE/UFRJ, p. 45.

Bird, S., Klein, E., Loper, E., 2009. Natural Language Processing with Python: Analyzing Text with the Natural Language Toolkit. O'Reilly Media, Inc., 2009.

Boeker, M., Vach, W., Motschall, E., 2013. Google scholar as replacement for systematic literature searches: good relative recall and precision are not enough. BMC Med. Res. Methodol. 13, 1–12.

Bourque, P., Dupuis, R., Abran, A., Moore, J.W., Tripp, L., 1999. The guide to the software engineering body of knowledge. IEEE Softw. 16 (6), 35–44.

Castañeda, V., Ballejos, L., Caliusco, M.L., Galli, M.R., 2010. The use of ontologies in requirements engineering. Glob. J. Res. Eng. 10 (6), 2–8.

Cunha, R.C., Cerqueira, C.S., 2023. Uma Proposta de Padrões de Sintaxe de Requisitos no Idioma Português.

Dermeval, D., et al., 2016. Applications of ontologies in requirements engineering: a systematic review of the literature. Requir. Eng. 21 (4), 405–437.

Di Maio, M., et al., 2021. Evaluating MBSE methodologies using the FEMMP framework. In: 2021 IEEE International Symposium on Systems Engineering. ISSE, pp. 1–8.

Easterbrook, S., Singer, J., Storey, M.-A., Damian, D., 2008. Selecting empirical methods for software engineering research. In: Guide to Advanced Empirical Software Engineering. pp. 285–311.

Estefan, J.A., et al., 2007. Survey of Model-Based Systems Engineering (MBSE) methodologies. Incose MBSE Focus. Group 25 (8), 1–12.

France, R., Evans, A., Lano, K., Rumpe, B., 1998. The UML as a formal modeling notation. Comput. Stand. Interfaces 19 (7), 325–334.
Friedenthal, S., Moore, A., Steiner, R., 2014. A Practical Guide to SysML: The Systems Modeling Language. Morgan Kaufmann.
Geogy, M., Dharani, A., 2016. A scrutiny of the software requirement engineering process. Procedia Technol. 25, 405–410.
Grady, J.O., 2009. Universal architecture description framework. Syst. Eng. 12 (2), 91–116.
Institute of Electrical and Eletronics Engineers, 1990. IEEE Std 610.12-1990: IEEE Standard Glossary of Software Engineering Terminology. IEEE.
Kitchenham, B., Brereton, O.P., Budgen, D., Turner, M., Bailey, J., Linkman, S., 2009. Systematic literature reviews in software engineering–a systematic literature review. Inf. Softw. Technol. 51 (1), 7–15.
Kitchenham, Barbara Ann, Budgen, David, Brereton, Pearl, 2015. Evidence-Based Software Engineering and Systematic Reviews, vol. 4, CRC Press.
Kitchenham, B., Charters, S., et al., 2007. Guidelines for Performing Systematic Literature Reviews in Software Engineering. UK.
Kossiakoff, A., Biemer, S.M., Seymour, S.J., Flanigan, D.A., 2020. Systems Engineering Principles and Practice. John Wiley & Sons.
Kotonya, G., Sommerville, I., 1998. Requirements Engineering: Processes and Techniques. Wiley Publishing.
Landis, J.R., Koch, G.G., 1977. The measurement of observer agreement for categorical data. Biometrics 159–174.
Loniewski, G., Insfran, E., Abrahão, S., 2010. A systematic review of the use of requirements engineering techniques in model-driven development. In: Model Driven Engineering Languages and Systems. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 213–227.
Matthew B, M., Michael, H.A., 1994. Qualitative data analysis. Sage Pub.
McGuinness, D.L., Van Harmelen, F., et al., 2004. OWL web ontology language overview. p. 2004, W3C recommendation, 10 (10).
Miller, G.A., 1995. WordNet: A lexical database for English. Commun. ACM 38 (11), 39–41.
Nicolás, J., Toval, A., 2009. On the generation of requirements specifications from software engineering models: A systematic literature review. Inf. Softw. Technol. 51 (9), 1291–1307.
Noguchi, R.A., 2016. Lessons Learned and Recommended Best Practices from Model-Based Systems Engineering (MBSE) Pilot Projects. Aerospace Corporation.
Nuseibeh, B., Easterbrook, S., 2000. Requirements engineering. In: Proceedings of the Conference on the Future of Software Engineering, Limerick Ireland.
OMG, 2019. Semantics of business vocabulary and rules version 1.5. [online] Available: https://www.omg.org/spec/SBVR/About-SBVR/.
Ramos, A.L., Ferreira, J.V., Barceló, J., 2011. Model-based systems engineering: An emerging approach for modern systems. IEEE Trans. Syst. Man, Cybern. Part C (Appl. Reviews) 42 (1), 101–111.
Rhodes, D., 2008. Addressing systems engineering challenges through collaborative research, SEARI—Systems engineering advancement research initiative.
Roques, P., 2017. Systems Architecture Modeling with the Arcadia Method: a Practical Guide to Capella. Elsevier.
Sànchez-Ferreres, J., Carmona, J., Padró, L., 2017. Aligning textual and graphical descriptions of processes through ILP techniques. In: Advanced Information Systems Engineering. Springer International Publishing, Cham, pp. 413–427.
Siegemund, K., Thomas, E.J., Zhao, Y., Pan, J., Assmann, U., 2011. Towards ontology-driven requirements engineering. In: Workshop Semantic Web Enabled Software Engineering At 10th International Semantic Web Conference. ISWC, Bonn.
Sommerville, I., 2011. Software Engineering, nineth ed. Addison-Wesley.
Stefanovic, D., Havzi, S., Nikolic, D., Dakic, D., Lolic, T., 2021. Analysis of the tools to support systematic literature review in software engineering. IOP Conf. Ser.: Mater. Sci. Eng. 1163, 012013.
Teufl, S., Khalil, M., Mou, D., 2013. Requirements for a model-based requirements engineering tool for embedded systems: Systematic literature review and survey. White paper. Fortiss GmbH.
Umar, M.A., Lano, K., 2024. Advances in automated support for requirements engineering: a systematic literature review. Requir. Eng..
Voirin, J.-L., 2018. Model-Based System and Architecture Engineering with the Arcadia Method. Elsevier.
Wohlin, C., 2014. Guidelines for snowballing in systematic literature studies and a replication in software engineering. In: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering. pp. 1–10.
Wohlin, C., Runeson, P., Höst, M., Ohlsson, M.C., Regnell, B., Wesslén, A., 2012. Experimentation in Software Engineering. Springer Science & Business Media.
Wolny, S., Mazak, A., Carpella, C., Geist, V., Wimmer, M., 2020. Thirteen years of SysML: a systematic mapping study. Softw. Syst. Model. 19, 111–169.
Yang, Z., Li, Z., Jin, Z., Chen, Y., 2014. A systematic literature review of requirements modeling and analysis for self-adaptive systems. In: Requirements Engineering: Foundation for Software Quality. Springer International Publishing, Cham, pp. 55–71.
Yue, T., Briand, L.C., Labiche, Y., 2011b. A systematic review of transformation approaches between user requirements and analysis models. Requir. Eng. 16 (2), 75–99.
Yue, T., Briand, L.C., Labiche, Y., 2015. Atoucan. ACM Trans. Softw. Eng. Methodol. 24 (3), 1–52.
Zhang, H., Babar, M.A., Tell, P., 2011. Identifying relevant studies in software engineering. Inform. Softw. Technol. 53 (6), 625–637.

Systematic literature review references

Abdelnabi, E.A., Maatuk, A.M., Abdelaziz, T.M., Elakeili, S.M., 2020. Generating UML class diagram using NLP techniques and heuristic rules. In: 2020 20th International Conference on Sciences and Techniques of Automatic Control and Computer Engineering. STA, Monastir, Tunisia.
Agt-Rickauer, H., Kutsche, R.-D., Sack, H., 2019. Automated recommendation of related model elements for domain models. In: Communications in Computer and Information Science. Springer International Publishing, Cham, pp. 134–158.
Alkhader, Y., Hudaib, A., Hammo, B., 2006. Experimenting with extracting software requirements using NLP approach. In: 2006 International Conference on Information and Automation, Colombo, Sri Lanka.
Arif, M., Mohammad, C.W., Sadiq, M., 2022. UML and NFR-framework based method for the analysis of the requirements of an information system. Int. J. Inf. Technol.
Arshad, S., Bajwa, I.S., Kazmi, R., 2019. Generating SBVR-XML representation of a controlled natural language. In: Communications in Computer and Information Science. Springer Singapore, Singapore, pp. 379–390.
Bajaj, D., Goel, A., Gupta, S.C., Batra, H., 2022. MUCE: a multilingual use case model extractor using GPT-3. Int. J. Inf. Technol. 14 (3), 1543–1554.
Bajwa, I.S., Bordbar, B., Anastasakis, K., Lee, M., 2012a. On a chain of transformations for generating alloy from NL constraints. In: Seventh International Conference on Digital Information Management. ICDIM 2012, Macau, Macao.
Bajwa, I.S., Choudhary, M.A., 2012. From natural language software specifications to UML class models. In: Enterprise Information Systems. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 224–237.
Bajwa, I.S., Lee, M., Bordbar, B., 2012b. Translating natural language constraints to OCL. J. King Saud Univ. - Comput. Inf. Sci. 24 (2), 117–128.
Bajwa, I.S., Siddique, M.I., Choudhary, M.A., 2007. Rule based production systems for automatic code generation in Java. In: 2006 1st International Conference on Digital Information Management.
Ballard, M., Peak, R., Cimtalay, S., Mavris, D., 2020. Bidirectional text-to-model element requirement transformation. In: 2020 IEEE Aerospace Conference, Big Sky, MT, USA.
Bao, T., Yang, J., Yang, Y., Yin, Y., 2022. RM2doc. In: Proceedings of the ACM/IEEE 44th International Conference on Software Engineering: Companion Proceedings, Pittsburgh Pennsylvania.
Bashir, N., Bilal, M., Liaqat, M., Marjani, M., Malik, N., Ali, M., 2021. Modeling class diagram using NLP in object-oriented designing. In: 2021 National Computing Colleges Conference. NCCC, Taif, Saudi Arabia.
Beimel, D., Kedmi-Shahar, E., 2019. Improving the identification of functional system requirements when novice analysts create use case diagrams: the benefits of applying conceptual mental models. Requir. Eng. 24 (4), 483–502.
Brace, W., Ekman, K., 2014. CORAMOD: a checklist-oriented model-based requirements analysis approach. Requir. Eng. 19 (1), 1–26.
Bryant, B.R., Lee, B.-S., 2003. Two-level grammar as an object-oriented requirements specification language. In: Proceedings of the 35th Annual Hawaii International Conference on System Sciences, Big Island, HI, USA.
Cardei, I., Fonoage, M., Shankar, R., 2008. Model based requirements specification and validation for component architectures. In: 2008 2nd Annual IEEE Systems Conference, Montreal, QC, Canada.
Carrillo, O., Chouali, S., Mountassir, H., 2014. Incremental modeling of system architecture satisfying SysML functional requirements. In: Formal Aspects of Component Software. Springer International Publishing, Cham, pp. 79–99.
Chen, J., Hu, B., Diao, W., Huang, Y., 2022. Automatic generation of SysML requirement models based on Chinese natural language requirements. In: Proceedings of the 2022 6th International Conference on Electronic Information Technology and Computer Engineering, Xiamen China.
Colombo, P., Khendek, F., Lavazza, L., 2011. Generating early design models from requirements analysis artifacts using problem frames and SysML. In: Modelling Foundations and Applications. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 97–114.
Colombo, P., Khendek, F., Lavazza, L., 2012. Bridging the gap between requirements and design: An approach based on problem frames and SysML. J. Syst. Softw. 85 (3), 717–745.
Dahhane, W., Zeaaraoui, A., Ettifouri, E.H., Bouchentouf, T., 2014. An automated object-based approach to transforming requirements to class diagrams. In: 2014 Second World Conference on Complex Systems. WCCS, Agadir, Morocco.
Danenas, P., Skersys, T., Butleris, R., 2020. Natural language processing-enhanced extraction of SBVR business vocabularies and business rules from UML use case diagrams. Data Knowl. Eng. 128 (101822), 101822.
Debnath, N., Leonardi, M., Mauco, M., Montejano, G., Riesco, D., 2008. Improving model driven architecture with requirements models. In: Fifth International Conference on Information Technology: New Generations. Itng 2008, Las Vegas, NV, USA.
Debnath, N., et al., 2011. An ATL transformation from natural language requirements models to business models of a MDA project. In: 2011 11th International Conference on ITS Telecommunications, St. Petersburg, Russia.
Deeptimahanti, D.K., Babar, M.A., 2009. An automated tool for generating UML models from natural language requirements. In: 2009 IEEE/ACM International Conference on Automated Software Engineering, Auckland, New Zealand.

Deeptimahanti, D.K., Sanyal, R., 2011. Semi-automatic generation of UML models from natural language requirements. In: Proceedings of the 4th India Software Engineering Conference, Thiruvananthapuram Kerala India.
Dong, Q., Wang, Z., Zhu, W., He, H., 2012. Capability requirements modeling and verification based on fuzzy ontology. J. Syst. Eng. Electron. 23 (1), 78–87.
Effa Bella, E., Creff, S., Gervais, M.-P., Bendraou, R., 2019. ATLaS: A framework for traceability links recovery combining information retrieval and semi-supervised techniques. In: 2019 IEEE 23rd International Enterprise Distributed Object Computing Conference. EDOC, Paris, France.
Elallaoui, M., Nafil, K., Touahni, R., 2018. Automatic transformation of user stories into UML use case diagrams using NLP techniques. Procedia Comput. Sci. 130, 42–49.
Elbendak, M., Vickers, P., Rossiter, N., 2011. Parsed use case descriptions as a basis for object-oriented class model generation. J. Syst. Softw. 84 (7), 1209–1223.
Fatwanto, A., Boughton, C., 2008. Analysis, specification and modeling of functional requirements for translative model-driven development. In: 2008 International Symposium on Knowledge Acquisition and Modeling, Wuhan, China.
Gulia, S., Choudhury, T., 2016. An efficient automated design to generate UML diagram from natural language specifications. In: 2016 6th International Conference - Cloud System and Big Data Engineering. Confluence, Noida, India.
Gupta, A., Poels, G., Bera, P., 2023. Generating multiple conceptual models from behavior-driven development scenarios. Data Knowl. Eng. 145 (102141), 102141.
Hamza, Z.A., Hammad, M., 2019. Generating UML use case models from software requirements using natural language processing. In: 2019 8th International Conference on Modeling Simulation and Applied Optimization. ICMSAO, Manama, Bahrain.
Harmain, H.M., Gaizauskas, R., 2000. CM-builder: an automated NL-based CASE tool. In: Proceedings ASE 2000. Fifteenth IEEE International Conference on Automated Software Engineering, Grenoble, France.
Ibrahim, M., Ahmad, R., 2010. Class diagram extraction from textual requirements using Natural Language Processing (NLP) techniques. In: 2010 Second International Conference on Computer Research and Development, Kuala Lumpur, Malaysia.
Iqbal, U., Bajwa, I.S., 2016. Generating UML activity diagram from SBVR rules. In: 2016 Sixth International Conference on Innovative Computing Technology. INTECH, Dublin, Ireland.
Iqbal, M., Rehman, A., 2019. Enrich exiting UML class model using ontology. In: Communications in Computer and Information Science. Springer Singapore, Singapore, pp. 573–578.
Jahan, M., Abad, Z.S.H., Far, B., 2021. Generating sequence diagram from natural language requirements. In: 2021 IEEE 29th International Requirements Engineering Conference Workshops. REW, Notre Dame, IN, USA.
Jura, J., Trnka, P., Cejnek, M., 2022. Using NLP to analyze requirements for agriculture 4.0 applications. In: 2022 23rd International Carpathian Control Conference. ICCC, Sinaia, Romania.
Jyothilakshmi, M.S., Samuel, P., 2012. Domain ontology based class diagram generation from functional requirements. In: 2012 12th International Conference on Intelligent Systems Design and Applications. ISDA, Kochi, India.
Kaundal, E.A., Kaur, A., 2017. CoMSS: Context based measure for semantic similarity between conceptual models. In: 2017 International Conference on Intelligent Computing and Control Systems. ICICCS, Madurai.
Kchaou, D., Bouassida, N., Mefteh, M., Ben-Abdallah, H., 2019. Recovering semantic traceability between requirements and design for change impact analysis. Innov. Syst. Softw. Eng. 15 (2), 101–115.
Klimek, R., 2013. From extraction of logical specifications to deduction-based formal verification of requirements models. In: Software Engineering and Formal Methods. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 61–75.
Ko, D., Kim, S., Park, S., 2019. Automatic recommendation to omitted steps in use case specification. Requir. Eng. 24 (4), 431–458.
Kornyshova, E., Deneckère, R., 2010. Decision-making ontology for information system engineering. In: Conceptual Modeling – ER 2010. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 104–117.
Krishnan, H., Samuel, P., 2010. Relative extraction methodology for class diagram generation using dependency graph. In: 2010 International Conference on Communication Control and Computing Technologies, Nagercoil, Tamil Nadu, India.
Kumar, D.D., Sanyal, R., 2008. Static UML model generator from analysis of requirements (SUGAR). In: 2008 Advanced Software Engineering and Its Applications, Hainan, China.
Leopold, H., Mendling, J., Polyvyanyy, A., 2012. Generating natural language texts from business process models. In: Notes on Numerical Fluid Mechanics and Multidisciplinary Design. Springer International Publishing, Cham, pp. 64–79.
Li, W., Hayes, J.H., Truszczyński, M., 2015a. Towards more efficient requirements formalization: A study. In: Requirements Engineering: Foundation for Software Quality. Springer International Publishing, Cham, pp. 181–197.
Li, F.-L., Horkoff, J., Borgida, A., Guizzardi, G., Liu, L., Mylopoulos, J., 2015b. From stakeholder requirements to formal specifications through refinement. In: Requirements Engineering: Foundation for Software Quality. Springer International Publishing, Cham, pp. 164–180.
Li, X., Liu, Z., 2008. Prototyping system requirements model. Electron. Notes Theor. Comput. Sci. (207), 17–32.
Li, X., Liu, Z., Schäf, M., Yin, L., 2010. AutoPA: Automatic prototyping from requirements. In: Lecture Notes in Computer Science, Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 609–624.
Lucassen, G., Robeer, M., Dalpiaz, F., van der Werf, J.M.E.M., Brinkkemper, S., 2017. EmphExtracting conceptual models from user stories with Visual Narrator. Requir. Eng. 22 (3), 339–358.
Maatuk, A.M., Abdelnabi, E.A., 2021. Generating UML use case and activity diagrams using NLP techniques and heuristics rules. In: International Conference on Data Science, E-Learning and Information Systems 2021, Ma'an Jordan.
Masuda, S., Matsuodani, T., Tsuda, K., 2016. Automatic generation of UTP models from requirements in natural language. In: 2016 IEEE Ninth International Conference on Software Testing, Verification and Validation Workshops. ICSTW, Chicago, IL, USA.
Mehmood, A., Mehmood, A., Akhtar, B., 2019. Enrich existing ontologies with new knowledge from existing artifacts. In: Communications in Computer and Information Science. Springer Singapore, Singapore, pp. 630–641.
Narawita, C.R., Vidanage, K., 2016. UML generator - an automated system for model driven development. In: 2016 Sixteenth International Conference on Advances in ICT for Emerging Regions. ICTer, Negombo, Sri Lanka.
Naumchev, A., Sadovykh, A., Ivanov, V., 2019. VERCORS: Hardware and software complex for intelligent round-trip formalized verification of dependable cyber–physical systems in a digital twin environment (position paper). In: Software Technology: Methods and Tools. Springer International Publishing, Cham, pp. 351–363.
Njonko, P.B.F., El Abed, W., 2012. From natural language business requirements to executable models via SBVR. In: 2012 International Conference on Systems and Informatics. ICSAI2012, Yantai, China.
Osman, M.S., Alabwaini, N.Z., Jaber, T.B., Alrawashdeh, T., 2019. Generate use case from the requirements written in a natural language using machine learning. In: 2019 IEEE Jordan International Joint Conference on Electrical Engineering and Information Technology. JEEIT, Amman, Jordan.
Patel, G.A., Priya, A.S., 2014. Resolve the uncertainity in requirement specification to generate the UML diagram. In: 2014 International Conference on Advances in Engineering and Technology. ICAET, Nagapattinam, India.
Pires, P.F., Delicato, F.C., Cóbe, R., Batista, T., Davis, J.G., Song, J.H., 2011. Integrating ontologies, model driven, and CNL in a multi-viewed approach for requirements engineering. Requir. Eng. 16 (2), 133–160.
Pröll, R., Rumpold, A., Bauer, B., 2018. Applying integrated domain-specific modeling for multi-concerns development of complex systems. In: Communications in Computer and Information Science. Springer International Publishing, Cham, pp. 247–271.
Rago, A., Abait, E., Marcos, C., Diaz-Pace, A., 2009. Early aspect identification from use cases using NLP and WSD techniques. In: Proceedings of the 15th Workshop on Early Aspects, Charlottesville Virginia USA.
Rago, A., Marcos, C., Diaz-Pace, J.A., 2016. Identifying duplicate functionality in textual use cases by aligning semantic actions. Softw. Syst. Model. 15 (2), 579–603.
Ramackers, G.J., Griffioen, P.P., Schouten, M.B.J., Chaudron, M.R.V., 2021. From prose to prototype: Synthesising executable UML models from natural language. In: 2021 ACM/IEEE International Conference on Model Driven Engineering Languages and Systems Companion. MODELS-C, Fukuoka, Japan.
Reis, A., da Silva, A.R., 2018. Evaluation of XIS-reverse, a model-driven reverse engineering approach for legacy information systems. In: Communications in Computer and Information Science. Springer International Publishing, Cham, pp. 23–46.
Ries, B., Capozucca, A., Guelfi, N., 2018. Messir: a text-first DSL-based approach for UML requirements engineering (tool demo). In: Proceedings of the 11th ACM SIGPLAN International Conference on Software Language Engineering, Boston MA USA.
Rivero, J.M., Grigera, J., Distante, D., Montero, F., Rossi, G., 2019. DataMock: An agile approach for building data models from user interface mockups. Softw. Syst. Model. 18 (1), 663–690.
Sajjad, R., Sarwar, N., 2016. NLP based verification of a UML class model. In: 2016 Sixth International Conference on Innovative Computing Technology. INTECH, Dublin, Ireland.
Sanyal, Shweta R., Ghoshal, B., 2018. Automatic extraction of structural model from semi structured software requirement specification. In: 2018 IEEE/ACIS 17th International Conference on Computer and Information Science. ICIS, Singapore.
Sarkar, S., Sharma, V.S., Agarwal, R., 2012. Creating design from requirements and use cases. In: Proceedings of the 5th India Software Engineering Conference, Kanpur India.
Sawant, K.P., Roy, S., Parachuri, D., Plesse, F., Bhattacharya, P., 2014. Enforcing structure on textual use cases via annotation models. In: Proceedings of the 7th India Software Engineering Conference, Chennai India.
Šenkýř, D., 2019. SHACL shapes generation from textual documents. In: Lecture Notes in Business Information Processing. Springer International Publishing, Cham, pp. 121–130.
Sharma, R., Gulia, S., Biswas, K., 2014. Automated generation of activity and sequence diagrams from natural language requirements. In: Proceedings of the 9th International Conference on Evaluation of Novel Approaches to Software Engineering, Lisbon, Portugal.
Sharma, R., Srivastava, P.K., Biswas, K.K., 2015. From natural language requirements to UML class diagrams. In: 2015 IEEE Second International Workshop on Artificial Intelligence for Requirements Engineering. AIRE, Ottawa, ON.

Shimada, H., Nakagawa, H., Tsuchiya, T., 2018. Constructing a goal model from requirements descriptions based on extraction rule. In: Communications in Computer and Information Science. Springer Singapore, Singapore, pp. 175–188.

Singh, S.K., Gupta, R., Sabharwal, S., Gupta, J.P., 2009a. Automatic extraction of events from textual requirements specification. In: 2009 World Congress on Nature & Biologically Inspired Computing. NaBIC, Coimbatore, India.

Singh, S.K., Sabharwal, S., Gupta, J.P., 2009b. E-XTRACT: A tool for extraction, analysis and classification of events from textual requirements. In: 2009 International Conference on Advances in Recent Technologies in Communication and Computing, Kottayam, Kerala, India.

Tahir, S., SarwarBajwa, I., 2013. Automatic event recognition and annotation in business requirement specifications. In: Eighth International Conference on Digital Information Management. ICDIM 2013, Islamabad, Pakistan.

Tichy, W.F., Koerner, S.J., 2010. Text to software. In: Proceedings of the FSE/SDP Workshop on Future of Software Engineering Research, Santa Fe New Mexico USA.

Tseng, F.S.C., Chen, C.-L., 2008. Enriching the class diagram concepts to capture natural language semantics for database access. Data Knowl. Eng. 67 (1), 1–29.

Tueno Fotso, S.J., Frappier, M., Laleau, R., Mammar, A., 2018. Modeling the hybrid ERTMS/ETCS level 3 standard using a formal requirements engineering approach. In: Lecture Notes in Computer Science. Springer International Publishing, Cham, pp. 262–276.

Umber, A., Bajwa, I.S., 2011. Minimizing ambiguity in natural language software requirements specification. In: 2011 Sixth International Conference on Digital Information Management, Melbourne, Australia.

Vasques, D.G., Santos, G.S., Gomes, F.D., Galindo, J.F., Martins, P.S., 2019. Use case extraction through knowledge acquisition. In: 2019 IEEE 10th Annual Information Technology, Electronics and Mobile Communication Conference. IEMCON, Vancouver, BC, Canada.

Wei, B., Delugach, H.S., Colmenares, E., Stringfellow, C., 2016. A conceptual graphs framework for teaching UML model-based requirements acquisition. In: 2016 IEEE 29th International Conference on Software Engineering Education and Training. CSEET, Dallas, TX, USA.

Wrycza, S., Marcinkowski, B., 2011. SysML requirement diagrams: Banking transactional platform case study. In: Research in Systems Analysis and Design: Models and Methods. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 15–22.

Yang, Z., et al., 2021. Exploiting augmented intelligence in the modeling of safety-critical autonomous systems. Form. Asp. Comput. 33 (3), 343–384.

Yue, T., Ali, S., 2012. Bridging the gap between requirements and aspect state machines to support non-functional testing: Industrial case studies. In: Modelling Foundations and Applications. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 133–145.

Yue, T., Ali, S., Briand, L., 2011a. Automated transition from use cases to UML state machines to support state-based testing. In: Modelling Foundations and Applications. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 115–131.

Yue, T., Briand, L.C., Labiche, Y., 2010. An automated approach to transform use cases into activity diagrams. In: Modelling Foundations and Applications. Springer Berlin Heidelberg, Berlin, Heidelberg, pp. 337–353.

Zeni, N., Kiyavitskaya, N., Mich, L., Cordy, J.R., Mylopoulos, J., 2015. GaiusT: supporting the extraction of rights and obligations for regulatory compliance. Requir. Eng. 20 (1), 1–22.

Zhang, H., Yue, T., Ali, S., Wu, J., Liu, C., 2017. A restricted natural language based use case modeling methodology for real-time systems. In: 2017 IEEE/ACM 9th International Workshop on Modelling in Software Engineering. MiSE, Buenos Aires, Argentina.

Zhong, S., Scarinci, A., Cicirello, A., 2023. Natural language processing for systems engineering: Automatic generation of systems modelling language diagrams. Knowl.-Based Syst. 259 (110071), 110071.

Zhu, R., Li, W., Jin, C., 2023. TAG: UML activity diagram deeply supervised generation from business textural specification. In: 2023 IEEE International Conference on Software Analysis, Evolution and Reengineering. SANER, Taipa, Macao.

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